Build Your First AI Chatbot Using Python

Summary:-

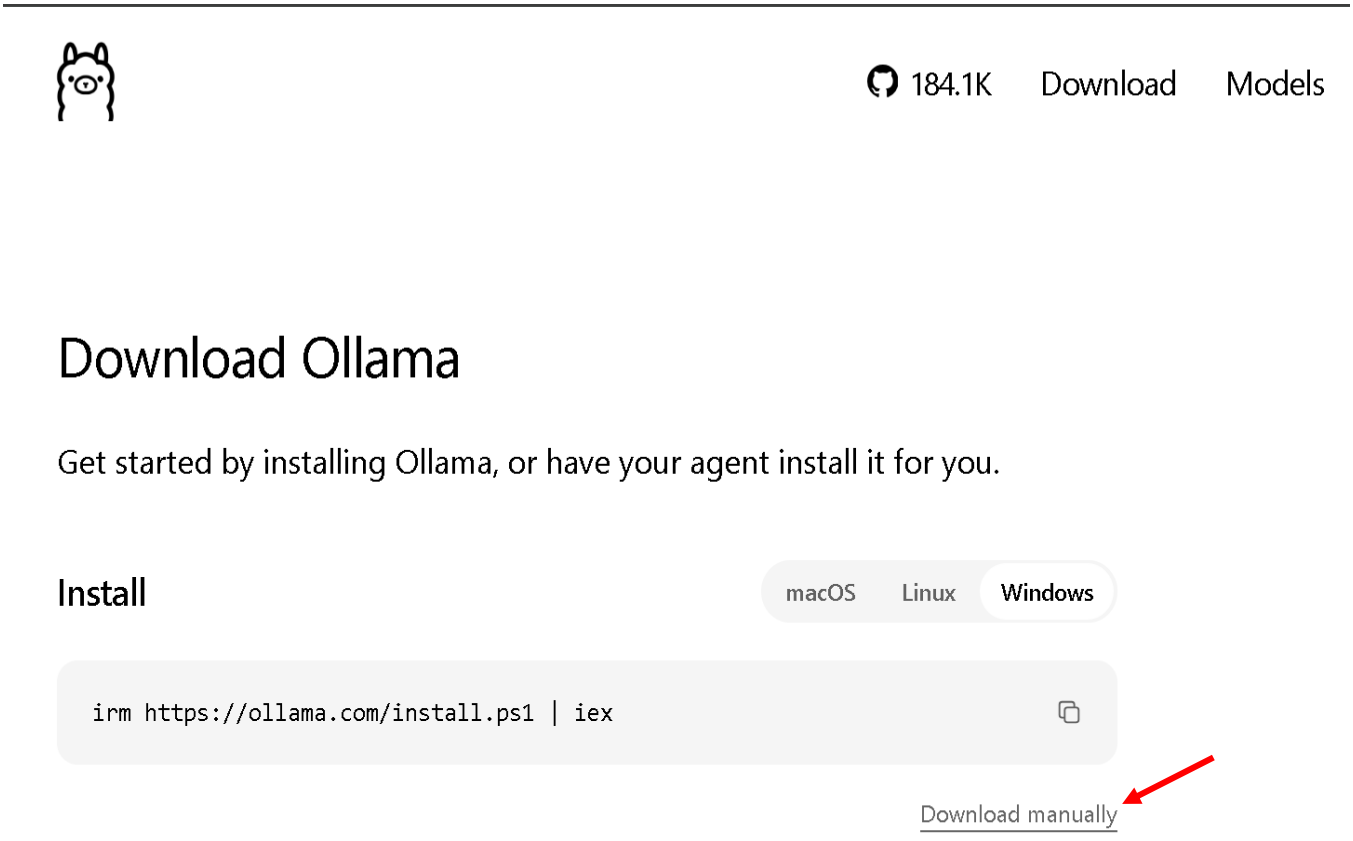
We created a basic AI chatbot using Python, Ollama, and the Qwen3:4B model. Using VS Code and the OpenAI Python library, we developed an interactive assistant that answers user questions through the terminal without requiring a paid cloud AI API.

To Build AI Applications we'll use free tools: -

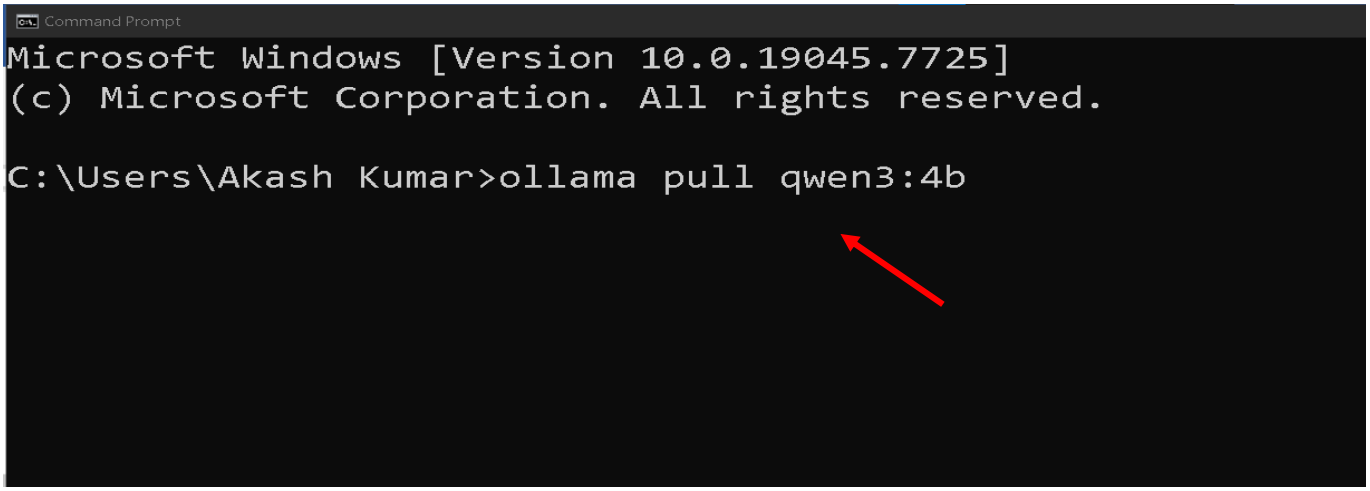
- Python
- Visual Studio Code
- Ollama
- Git

Lets Start: -

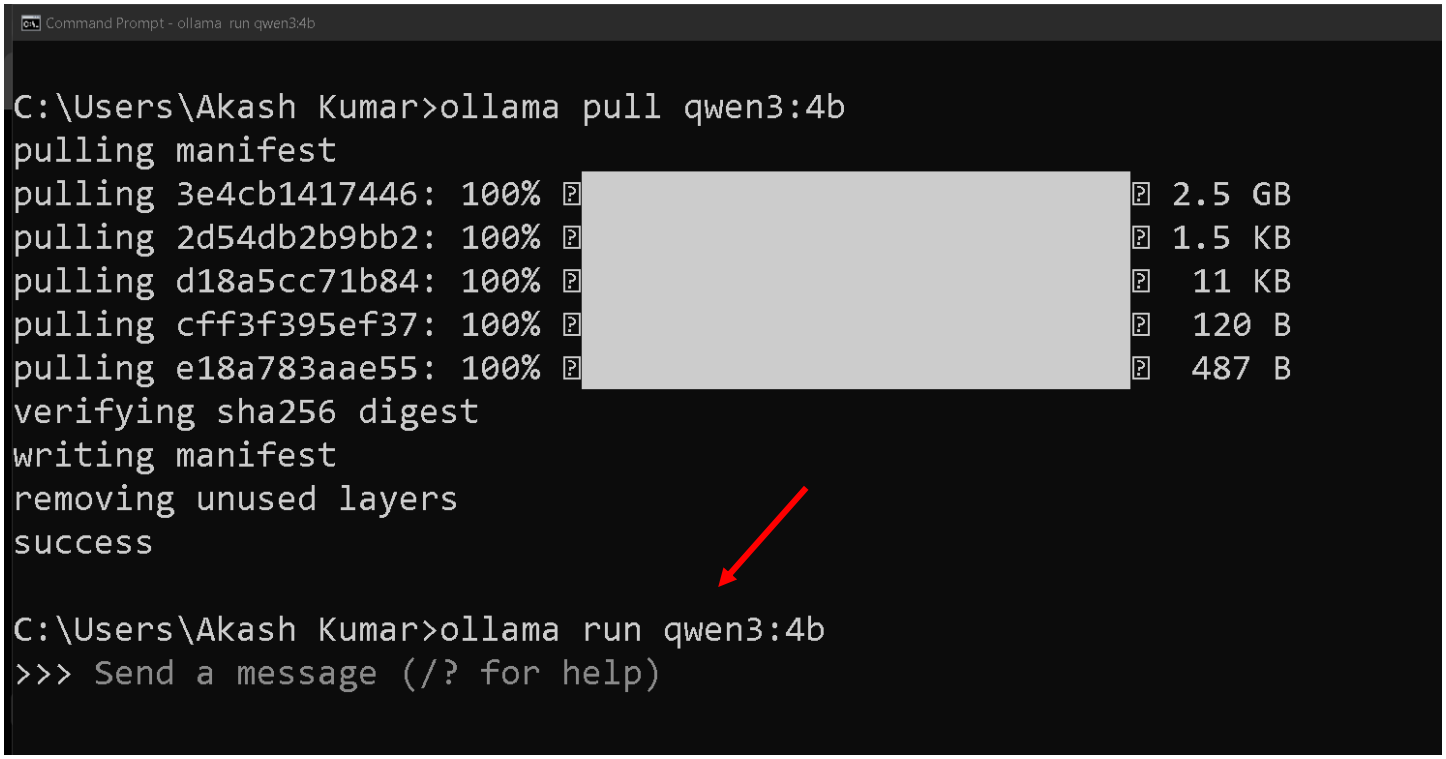
- First download Ollama - Link



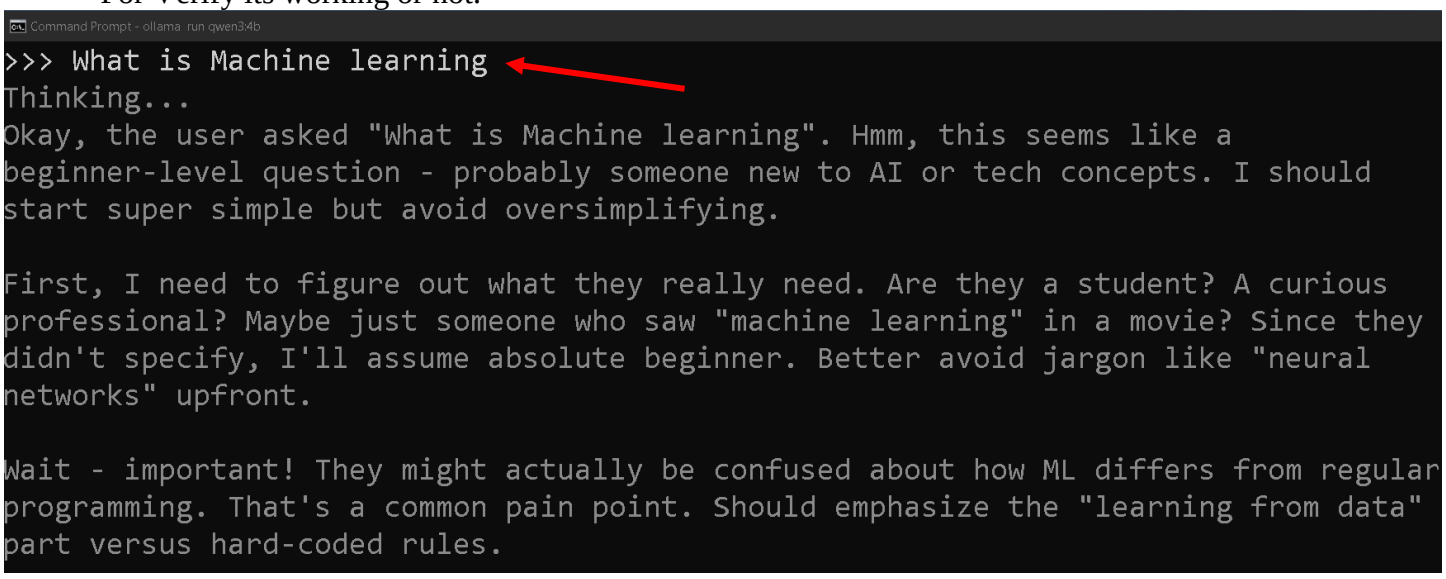
- After downloading Install Ollama.
- Then After installation open command prompt.



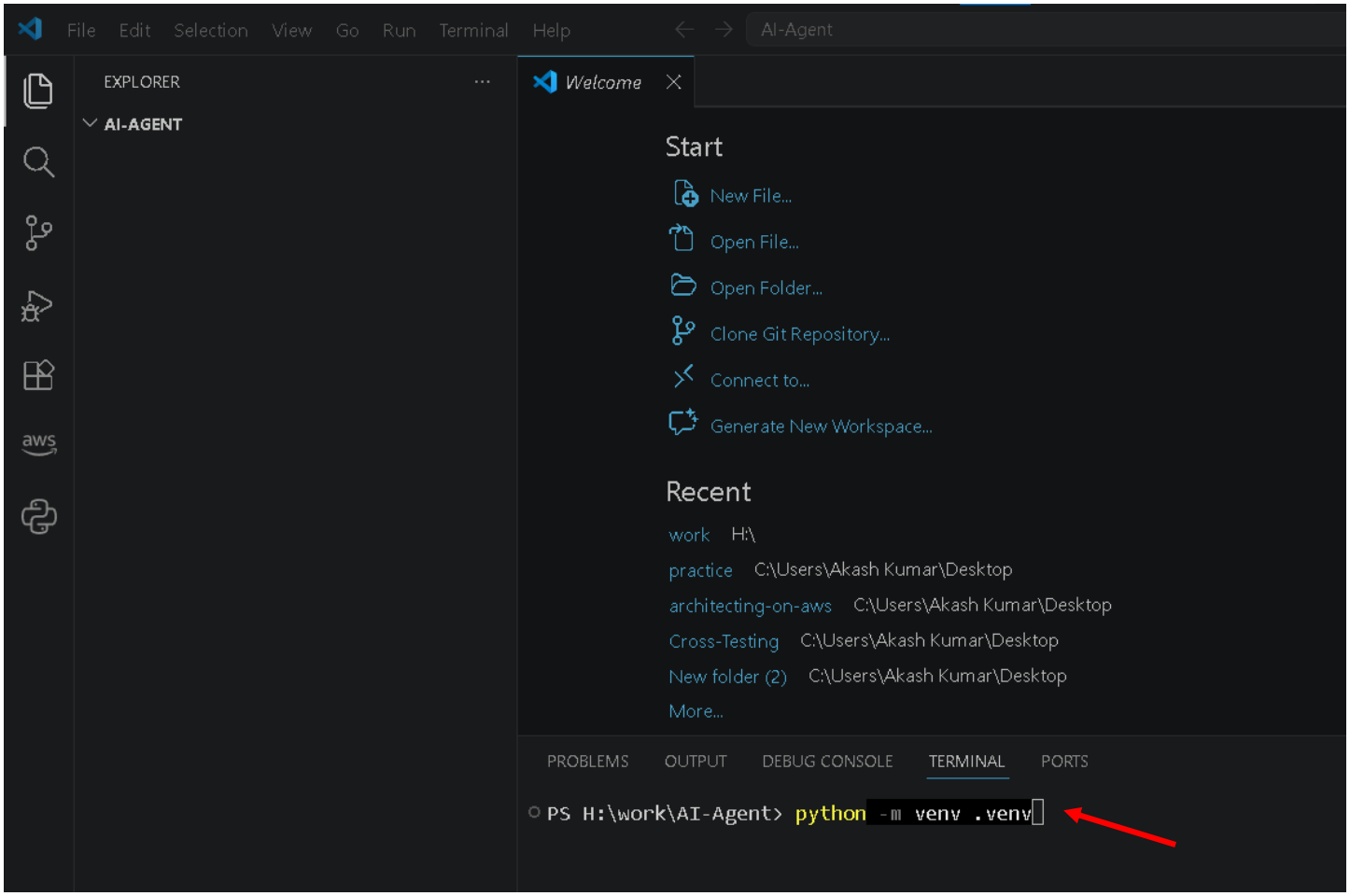
- And Enter ollama pull qwen3:4b
- It will take time.



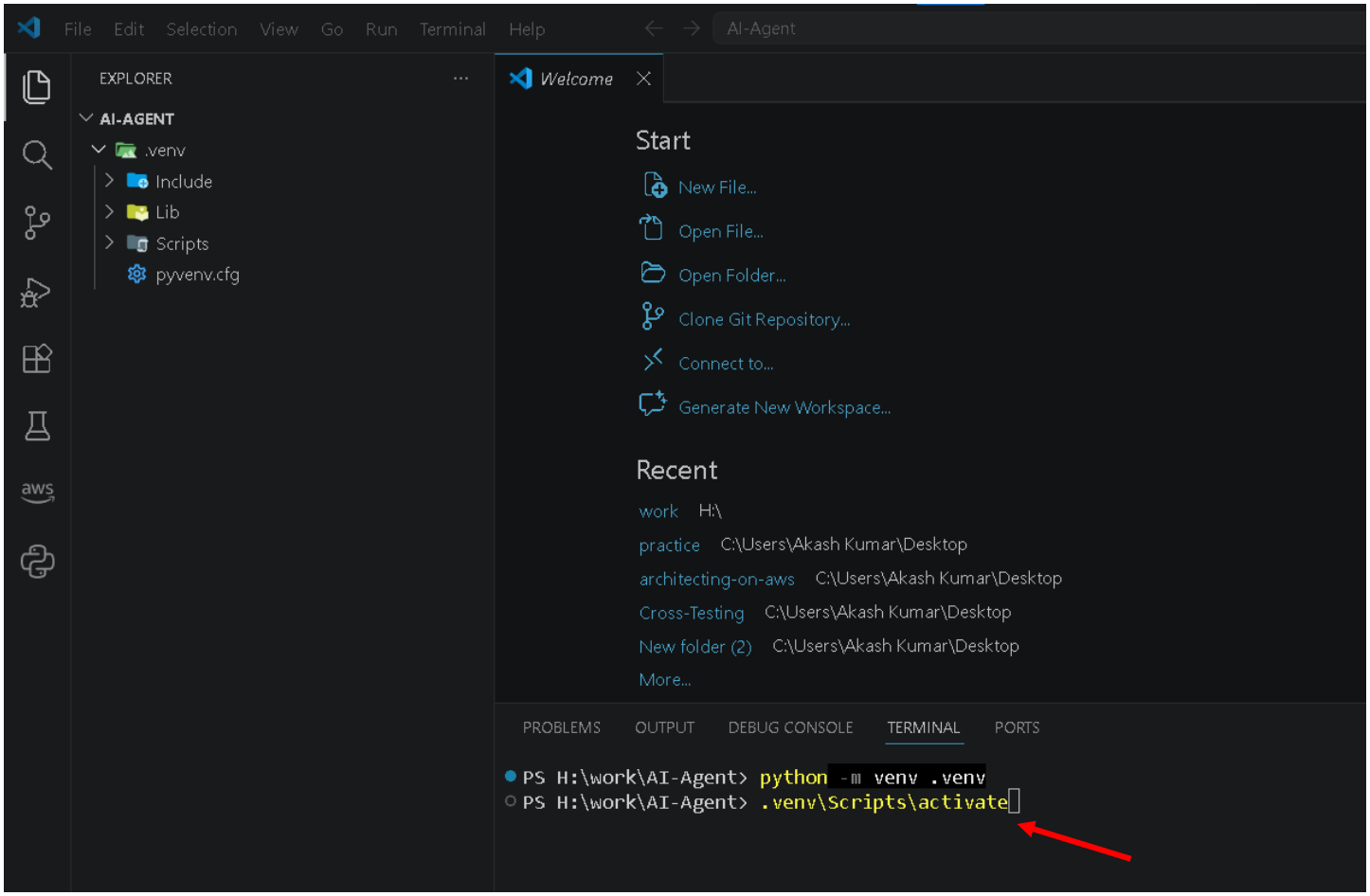
- Now after Downloading AI model.
- Enter ollama run qwen3:4b
- For Verify its working or not.



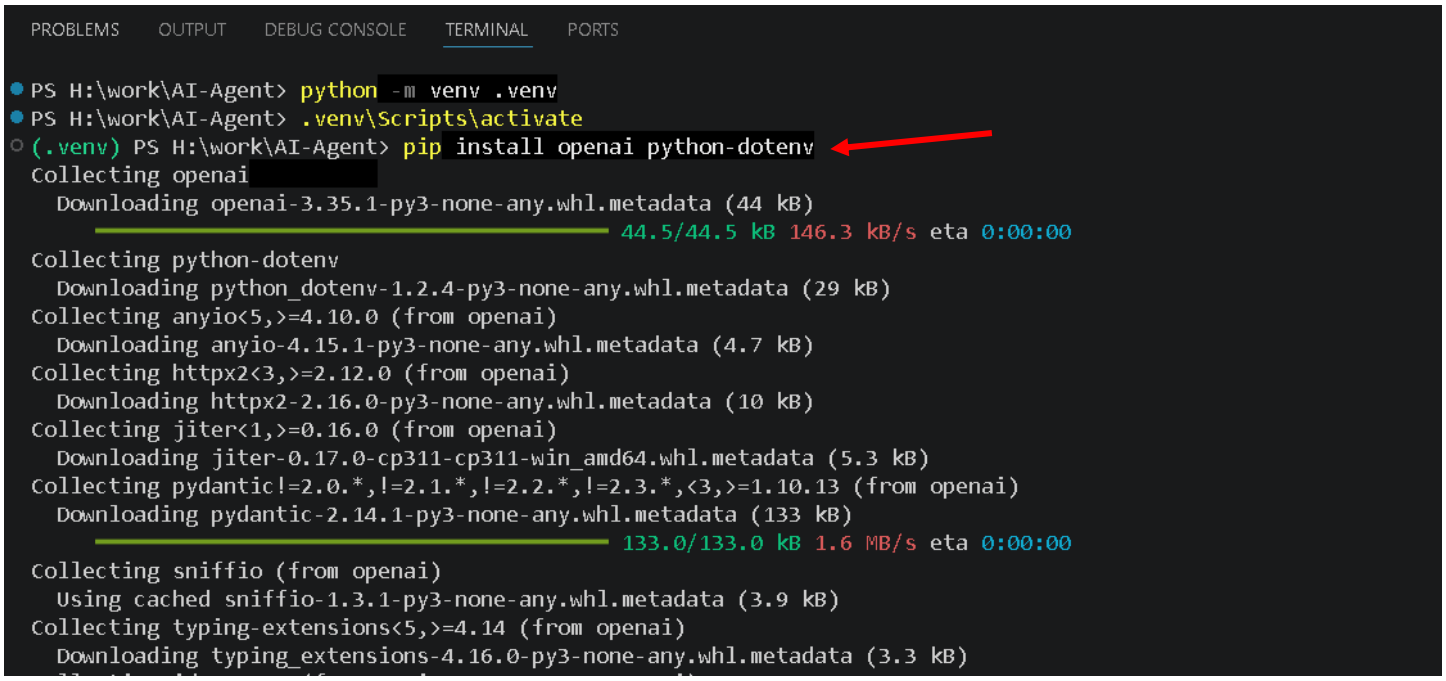
- ask some questions and check its work properly.
- Now create your project folder in your system and open in Vs Code.



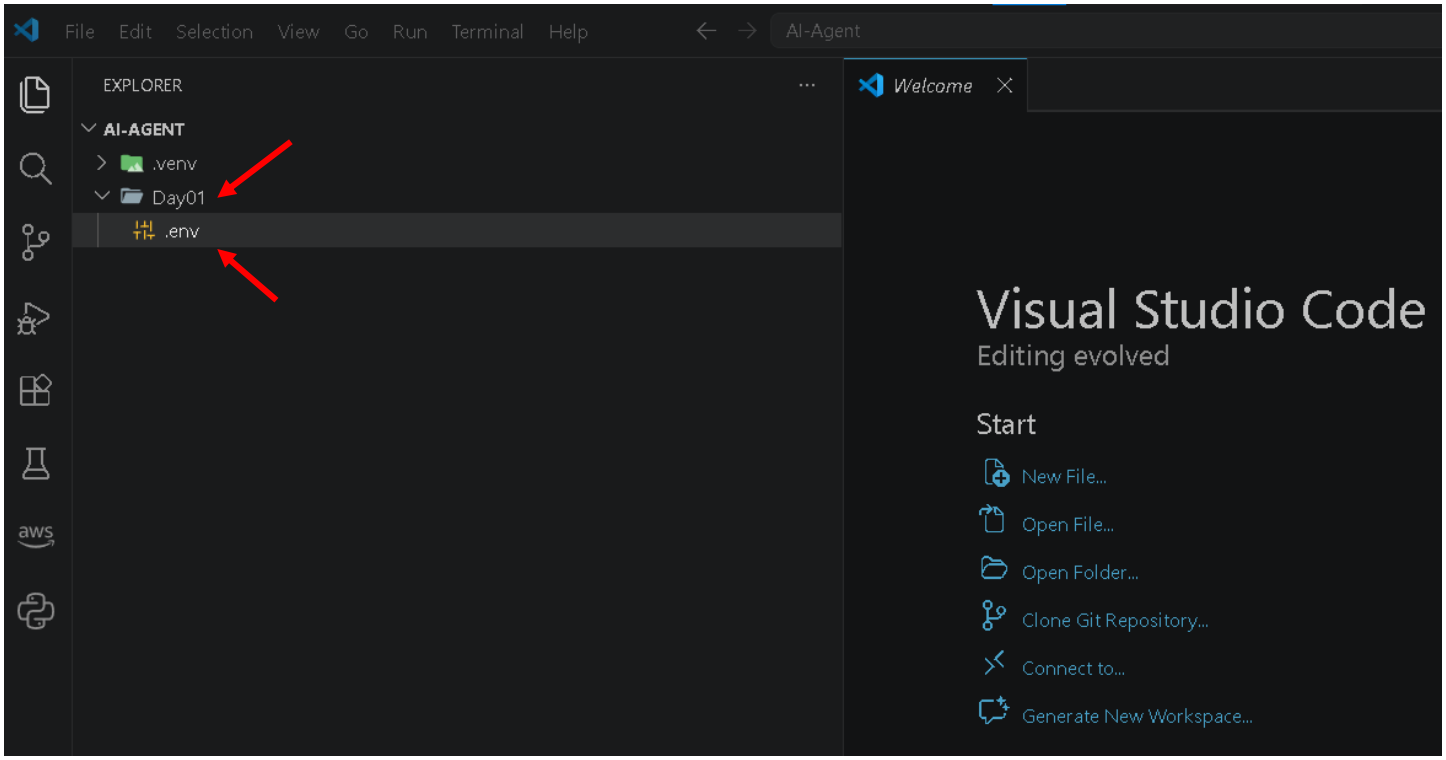
- Our Folder AI-AGENT is opened in Vs Code.
- Now we need to create virtual environment .
- So open terminal and Enter python -m venv .venv



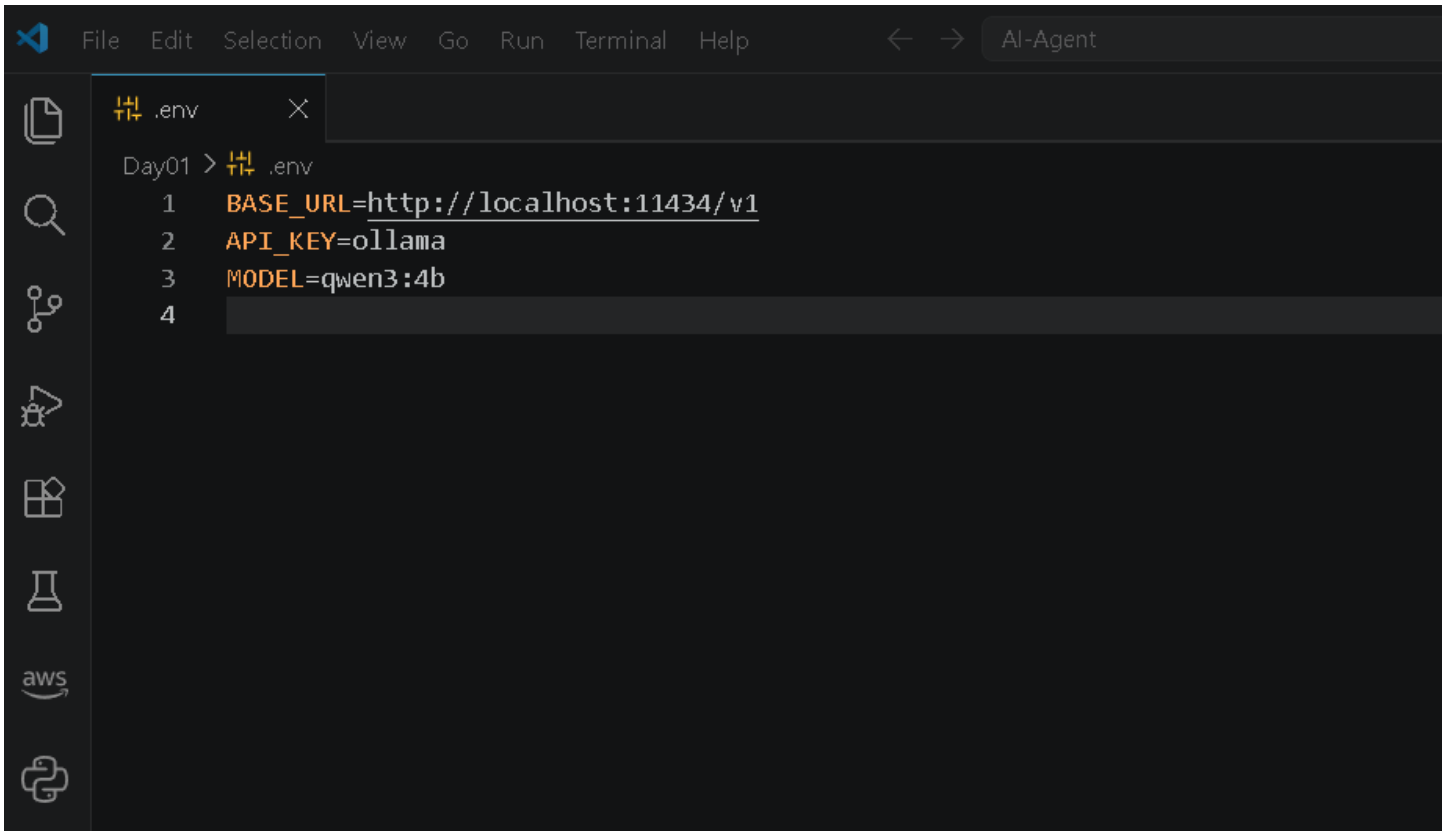
- Now when virtual environment is created we need to activate,
- So enter .venv\Scripts\activate
- Now we need to install some python Library.



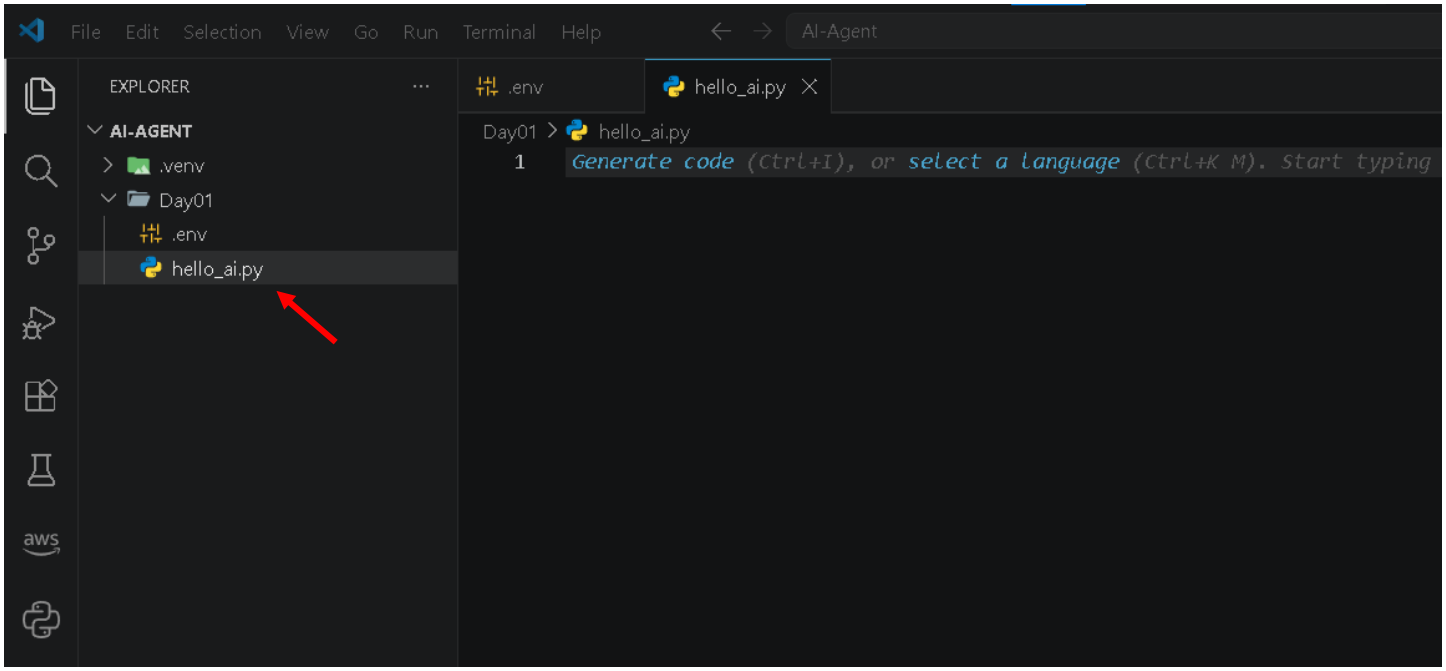
- So We Enter pip install openai python-dotenv



- Now After installation create folder inside AI-AGENT.
- And inside new folder create a new file .env



- Now open .env file and Enter these things:-
- BASE_URL=http://localhost:11434/v1
- API_KEY=ollama
- MODEL=qwen3:4b



- Now create New file in same folder hello_ai.py
- And Enter the Following codes: -

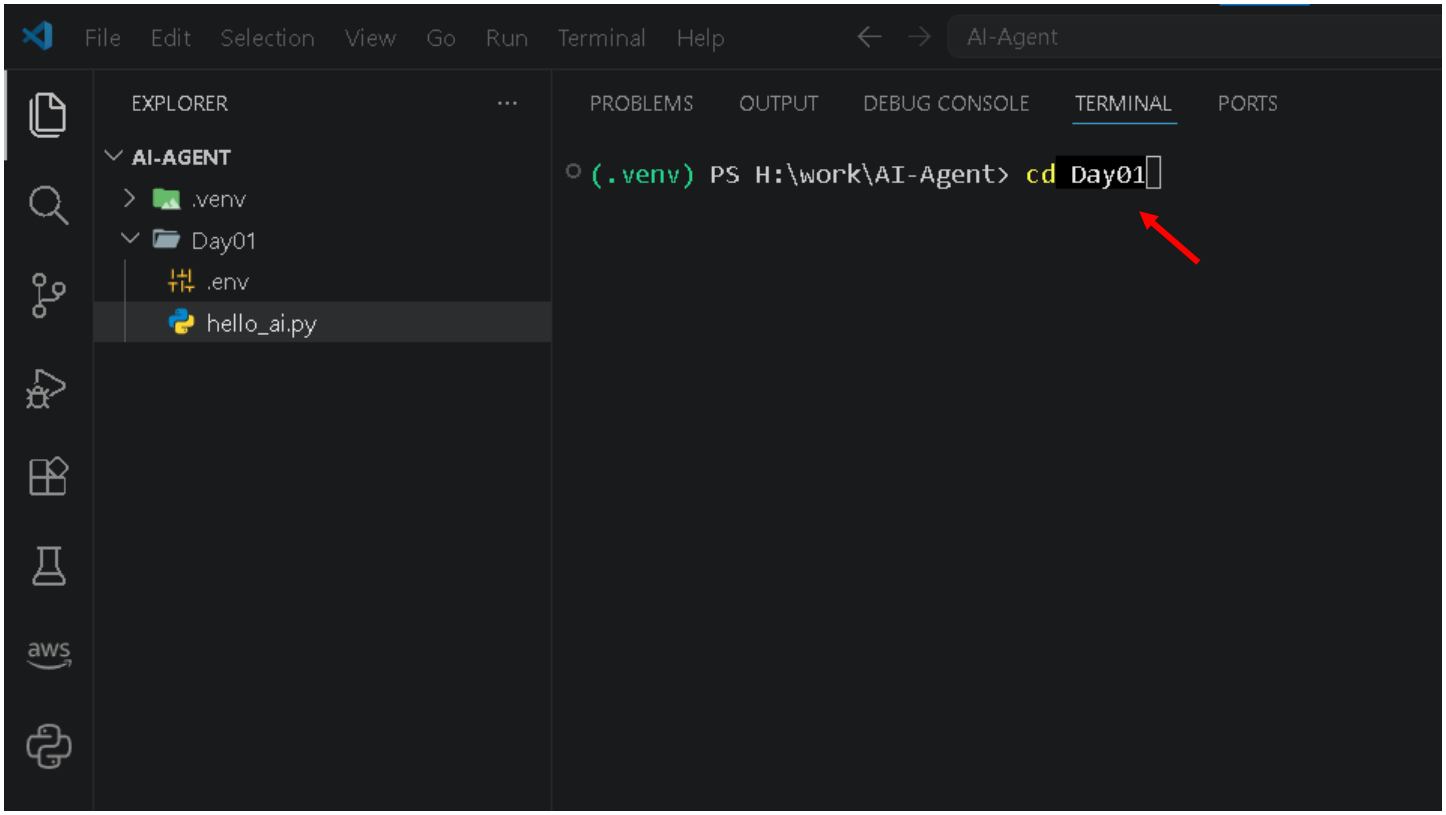
```
from openai import OpenAI
from dotenv import load_dotenv
import os

# Load configuration from the .env file
load_dotenv()

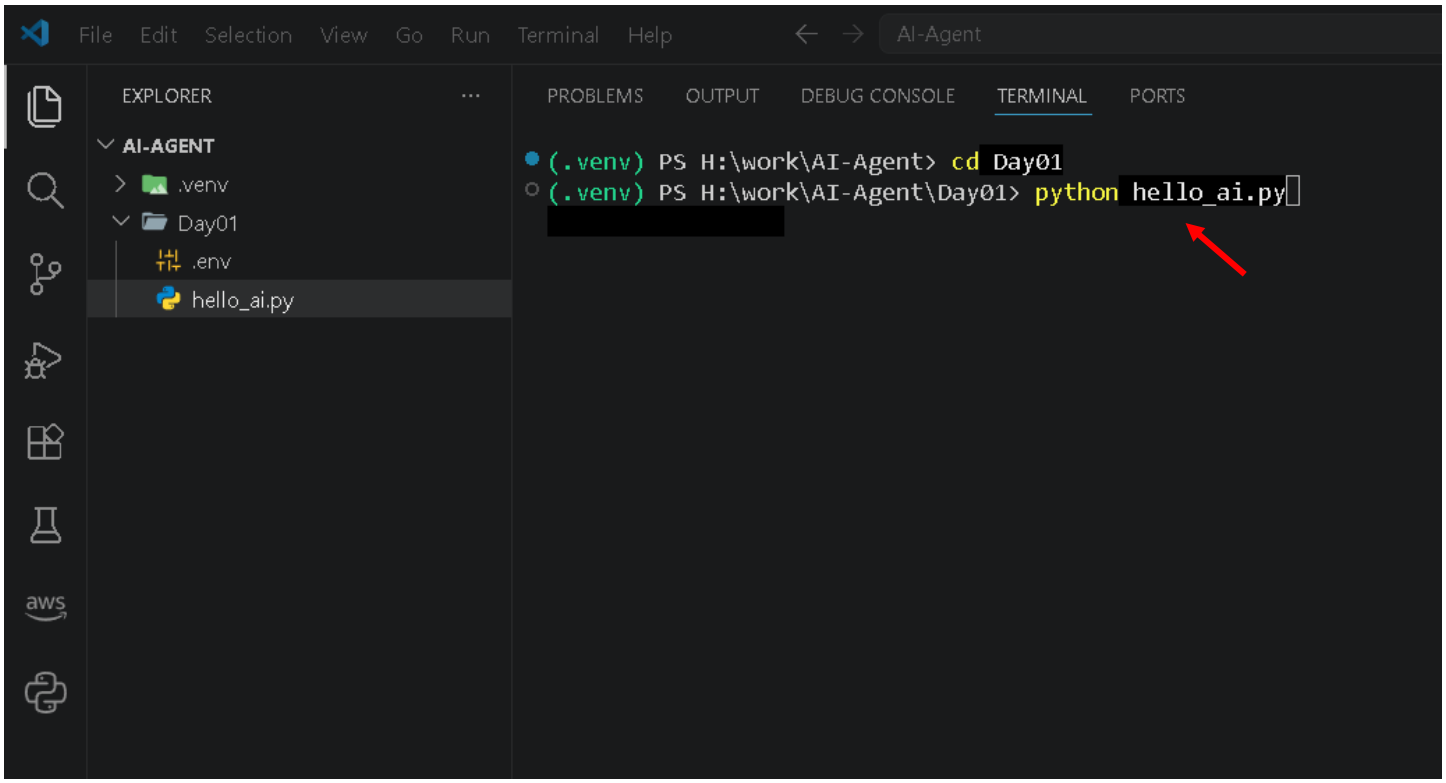
# Create a client that communicates with Ollama
client = OpenAI(
    base_url=os.getenv("BASE_URL"),
    api_key=os.getenv("API_KEY")
)

# Send a question to the AI model
response = client.chat.completions.create(
    model=os.getenv("MODEL"),
    messages=[
        {
            "role": "user",
            "content": "What is Artificial Intelligence?"
        }
    ]
)

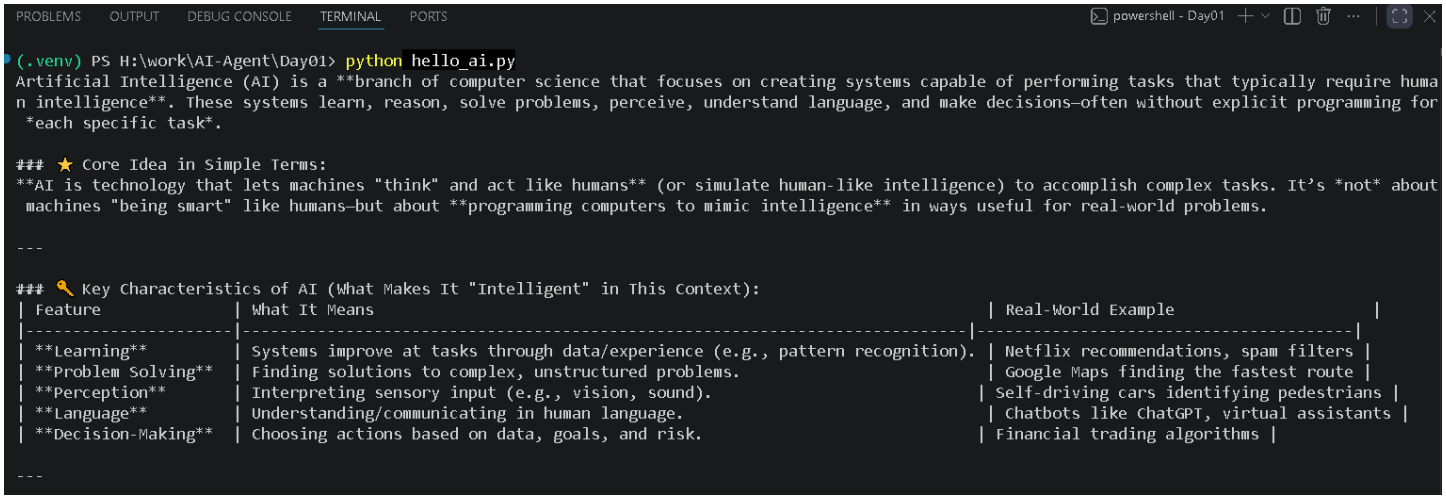
# Display the response
print(response.choices[0].message.content)
```



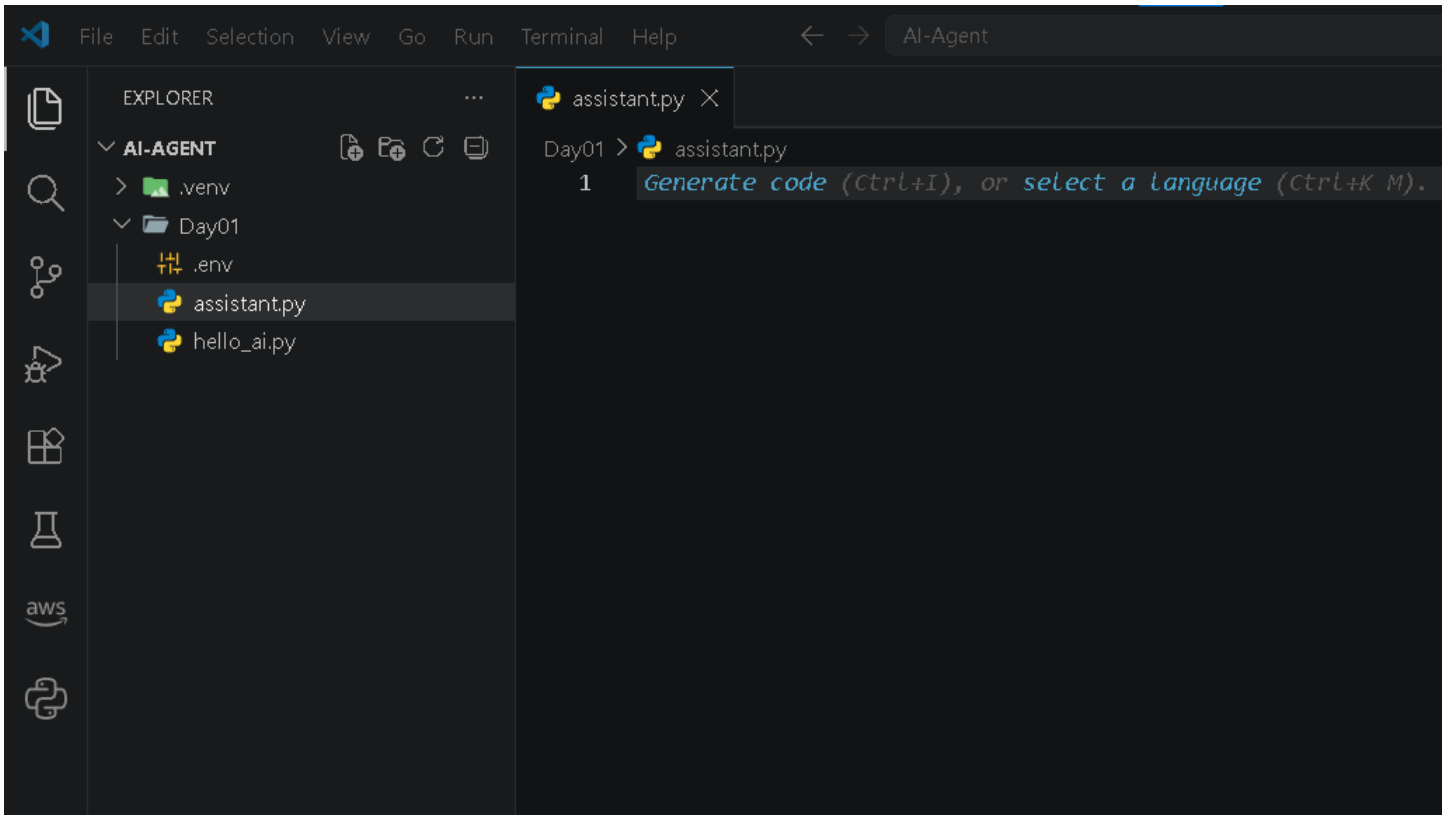
- Now open the terminal and go inside Day01 directory.
- So for this enter cd Day01 in terminal.



- Now Enter python hello_ai.py



- Now see it will return output according our given content in our code.
- But one issue is that if we need to chat about new topic we need to go inside code and change content.
- So now we can resolve it.



- Now we create a new python file assistant.py
- Now enter the following code: -

```
from openai import OpenAI
from dotenv import load_dotenv
import os

# Load configuration
load_dotenv()

# Create AI client
client = OpenAI(
    base_url=os.getenv("BASE_URL"),
    api_key=os.getenv("API_KEY")
)
```

```
print("=" * 40)
print("      My AI Assistant")
print("=" * 40)

while True:

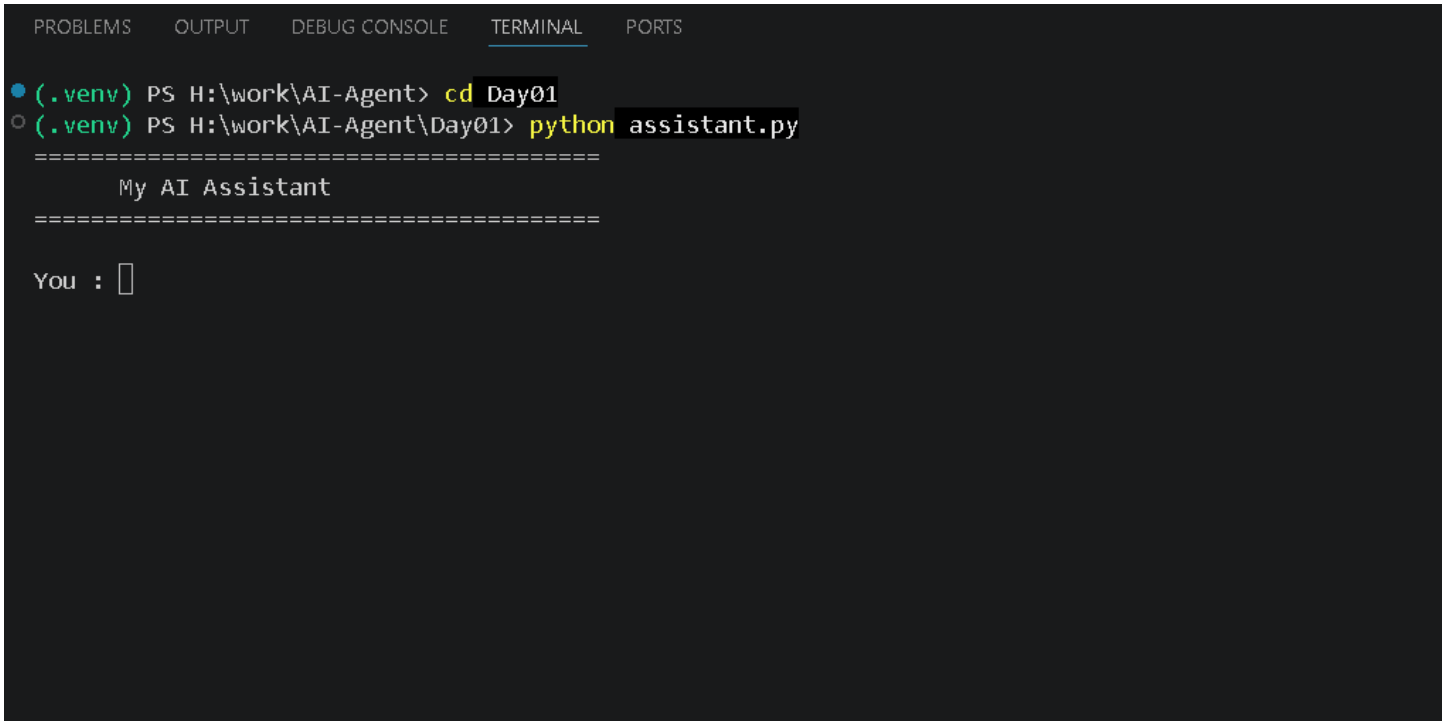
    user_input = input("\nYou : ")

    if user_input.lower() == "quit":
        print("\nAI  : Goodbye! Have a great day.")
        break

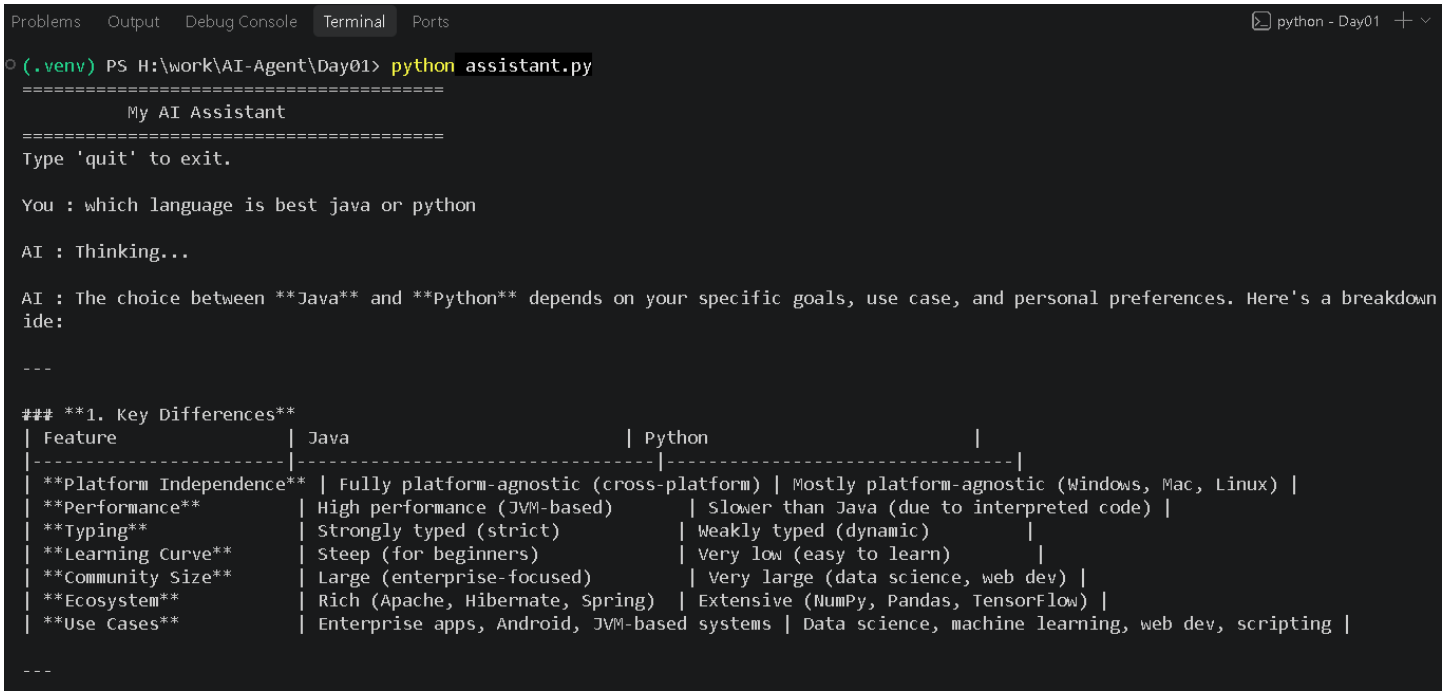
    response = client.chat.completions.create(
        model=os.getenv("MODEL"),
        messages=[
            {
                "role": "user",
                "content": user_input
            }
        ]
    )

    print("\nAI :", response.choices[0].message.content)
```

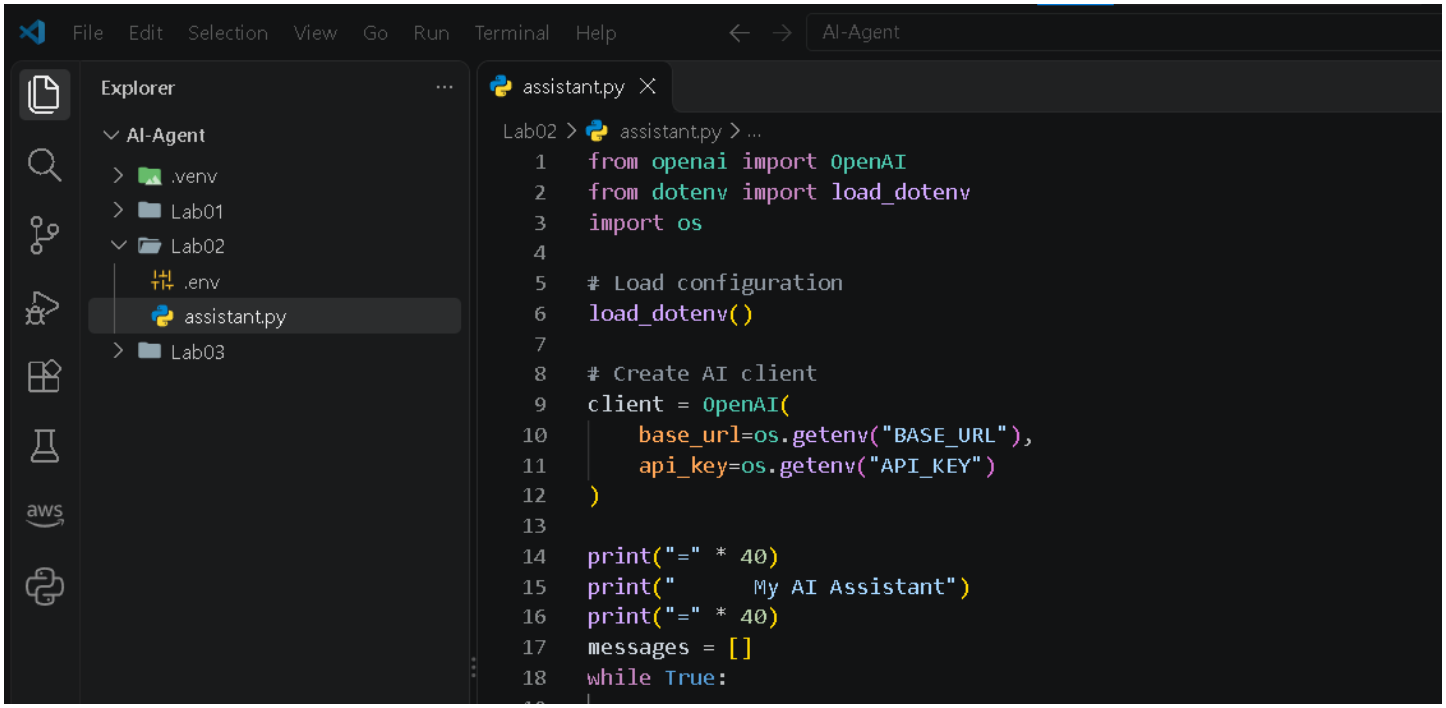
- Now open terminal.



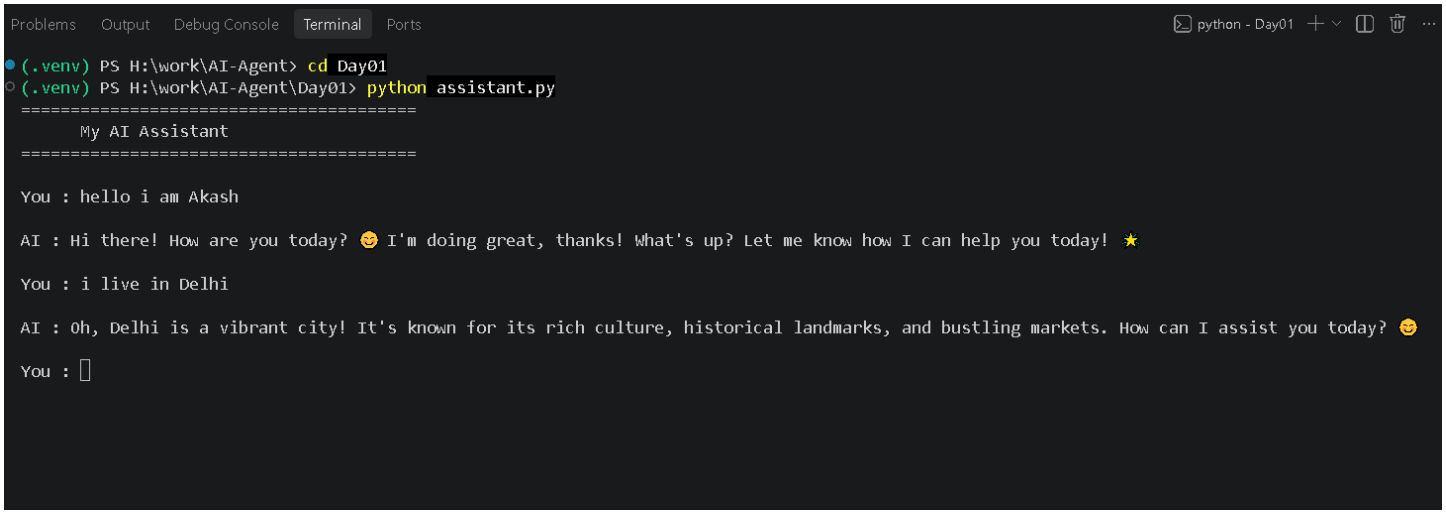
- And Enter python assistant.py
- Now see our assistant ask topic in our terminal not in code.
- Now check its work perfectly.



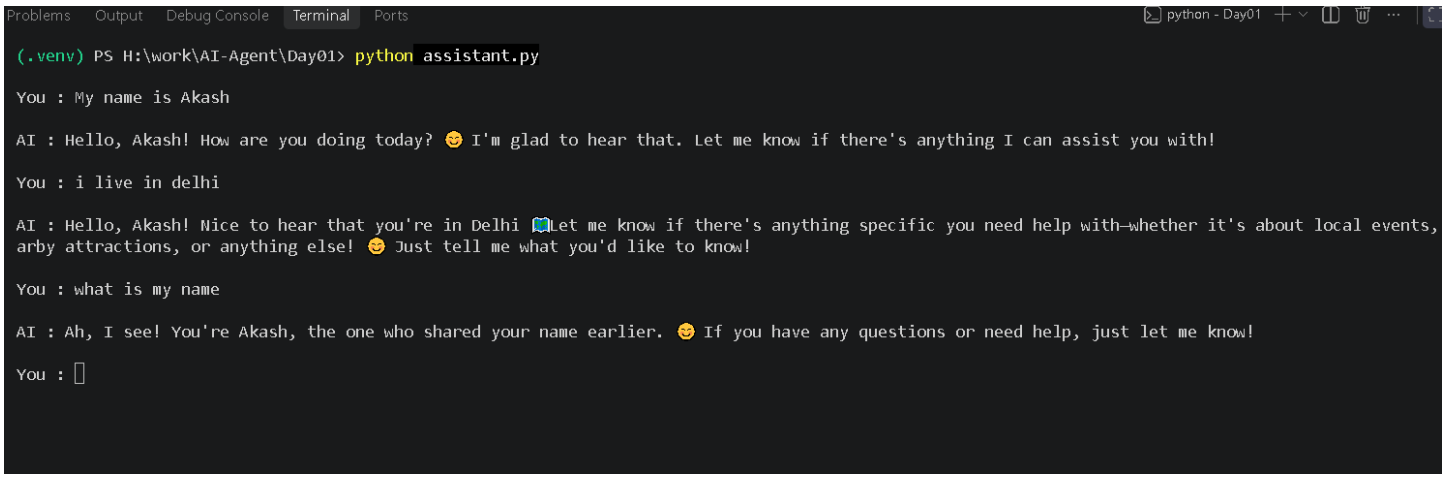
- Now you see I ask which language is best java or python.
- Out chat bot reply perfectly.
- Now we need to provide memory to our assistant.



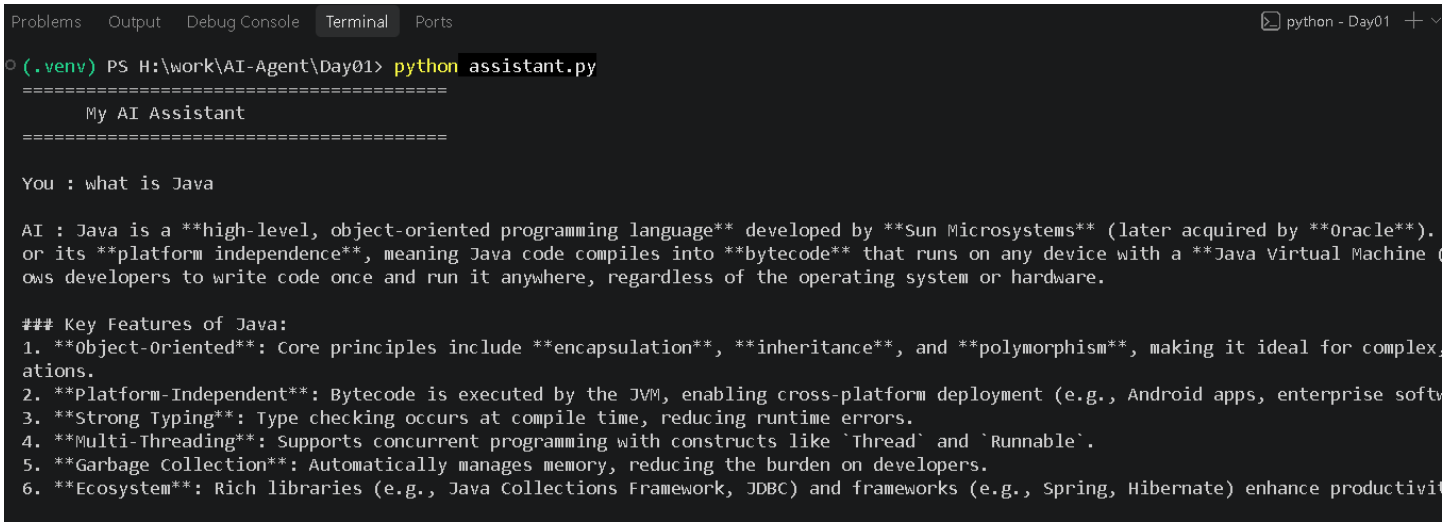
- Now we create a folder Lab02 where we work on Memory part.
- So we update our code assistant.py
- Code is provided in resource.



- After updating my assistant.py code
- I give some input about me.
- Now let's see it work correctly and give output.



- Now see I ask what is my name it tells my name.
- Now we need to add a conversation memory.
- It means our agent remember our previous chats.
- So we update our assistant.py code.
- Updated code is provided in resource.



- Now see I ask what is java then it provides output.
- And when you scroll down you see conversation history.

```
Problems  Output  Debug Console  Terminal  Ports                                    python - Day01

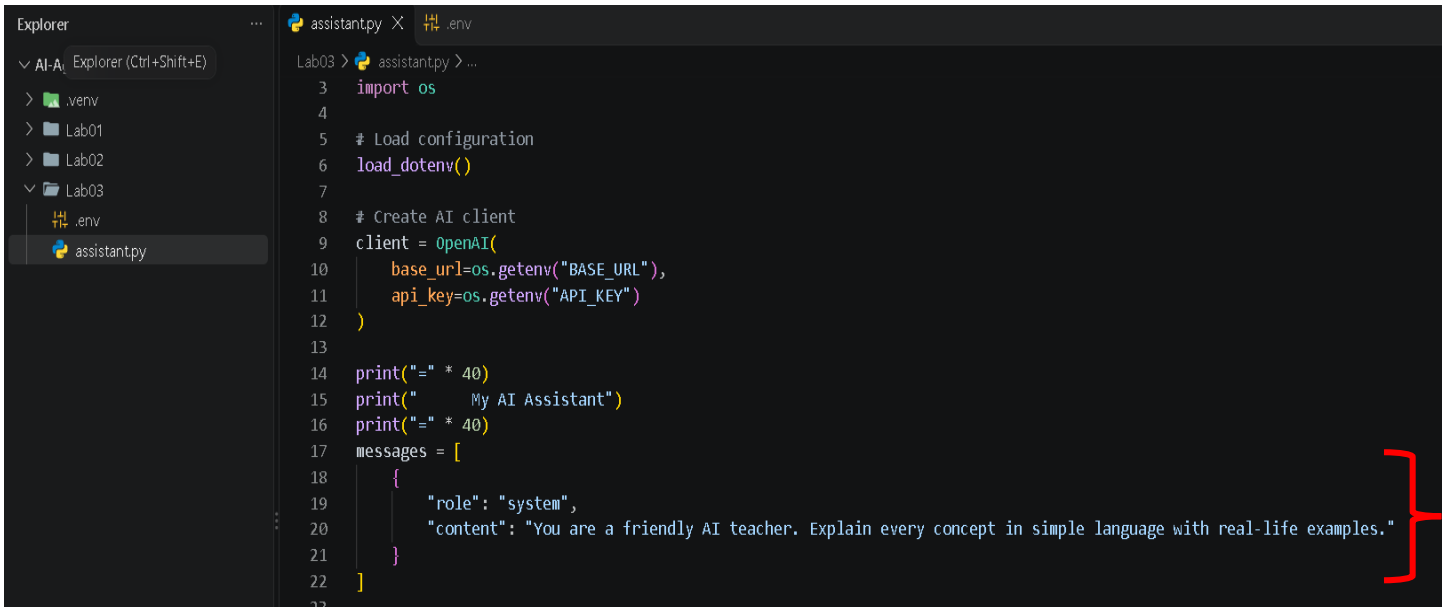
(.venv) PS H:\work\AI-Agent\Day01> python assistant.py

========== Conversation History ==========
User : what is Java
Assistant : Java is a **high-level, object-oriented programming language** developed by **Sun Microsystems** (later acquired by **Oracle**). It is ren
owned for its **platform independence**, meaning Java code compiles into **bytecode** that runs on any device with a **Java Virtual Machine (JVM)**. T
his allows developers to write code once and run it anywhere, regardless of the operating system or hardware.

### Key Features of Java:
1. **Object-Oriented**: Core principles include **encapsulation**, **inheritance**, and **polymorphism**, making it ideal for complex, scalable applic
ations.
2. **Platform-Independent**: Bytecode is executed by the JVM, enabling cross-platform deployment (e.g., Android apps, enterprise software).
3. **Strong Typing**: Type checking occurs at compile time, reducing runtime errors.
4. **Multi-Threading**: Supports concurrent programming with constructs like `Thread` and `Runnable`.
5. **Garbage Collection**: Automatically manages memory, reducing the burden on developers.
6. **Ecosystem**: Rich libraries (e.g., Java Collections Framework, JDBC) and frameworks (e.g., Spring, Hibernate) enhance productivity.

### Use Cases:
- **Android Development**: Android apps are primarily built using Java (though Kotlin is now the preferred language).
- **Enterprise Software**: Used for building robust, scalable applications.
- **Web Applications**: Java is used in frameworks like **Java Servlets** and **Java Spring Boot**.
- **Desktop Applications**: Java Swing and JavaFX for graphical user interfaces.
```

- Basically this conversation history is for debugging process.
- So our memory work is done.
- Now we work on our Assistant Personality on Lab03.



- Now we blank message in assistant.py
- And add Message with role (System) and content (what you want that your assistant behave like).
- In my scenario I give content (You are a friendly AI teacher. Explain every concept in simple language with real-life examples)
- Code is provided in resource.
- Now we see how our assistant behave if we give topic.

```
Problems  Output  Debug Console  Terminal  Ports                                                                    python

(.venv) PS H:\work\AI-Agent> cd Lab03
(.venv) PS H:\work\AI-Agent\Lab03> python assistant.py
========================================
      My AI Assistant
========================================

You : Explain loops

AI : Sure! Let me explain loops with a **simple, relatable example**! 😊

---

### What is a Loop?
A **loop** is like a **robot** that does the same thing **over and over** until something happens. For example, if you want to **water your plants eve
ry day**, you can use a loop to automatically do that without doing it manually each day!

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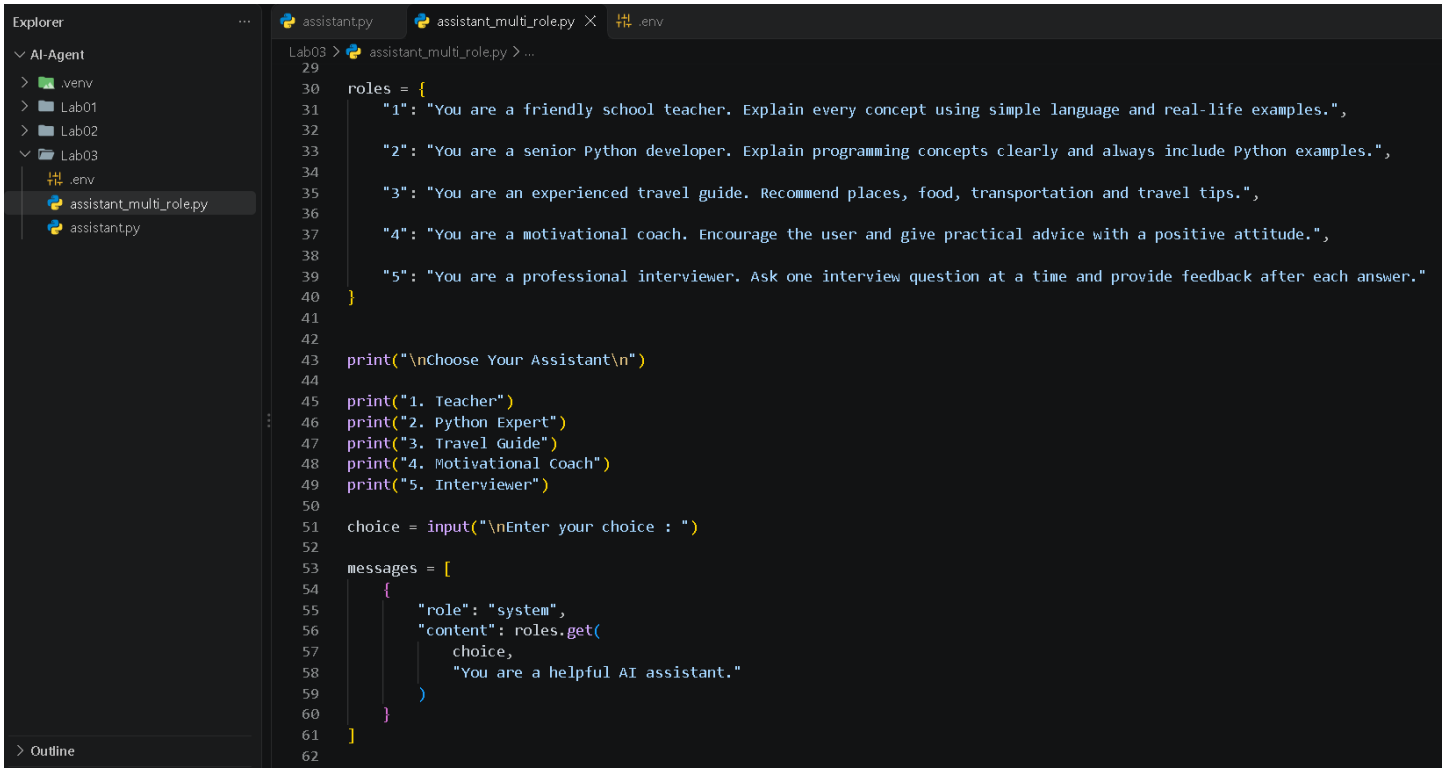
### Types of Loops
There are two main types of loops:

#### 1. **For Loop**
Think of it as a "counting" loop. It repeats a fixed number of times.
**Example**:
If you want to add numbers from 1 to 10, you can use a loop that runs **10 times**.
- Step 1: Start at 1
- Step 2: Add 1 to the total
- Step 3: Repeat until you reach 10

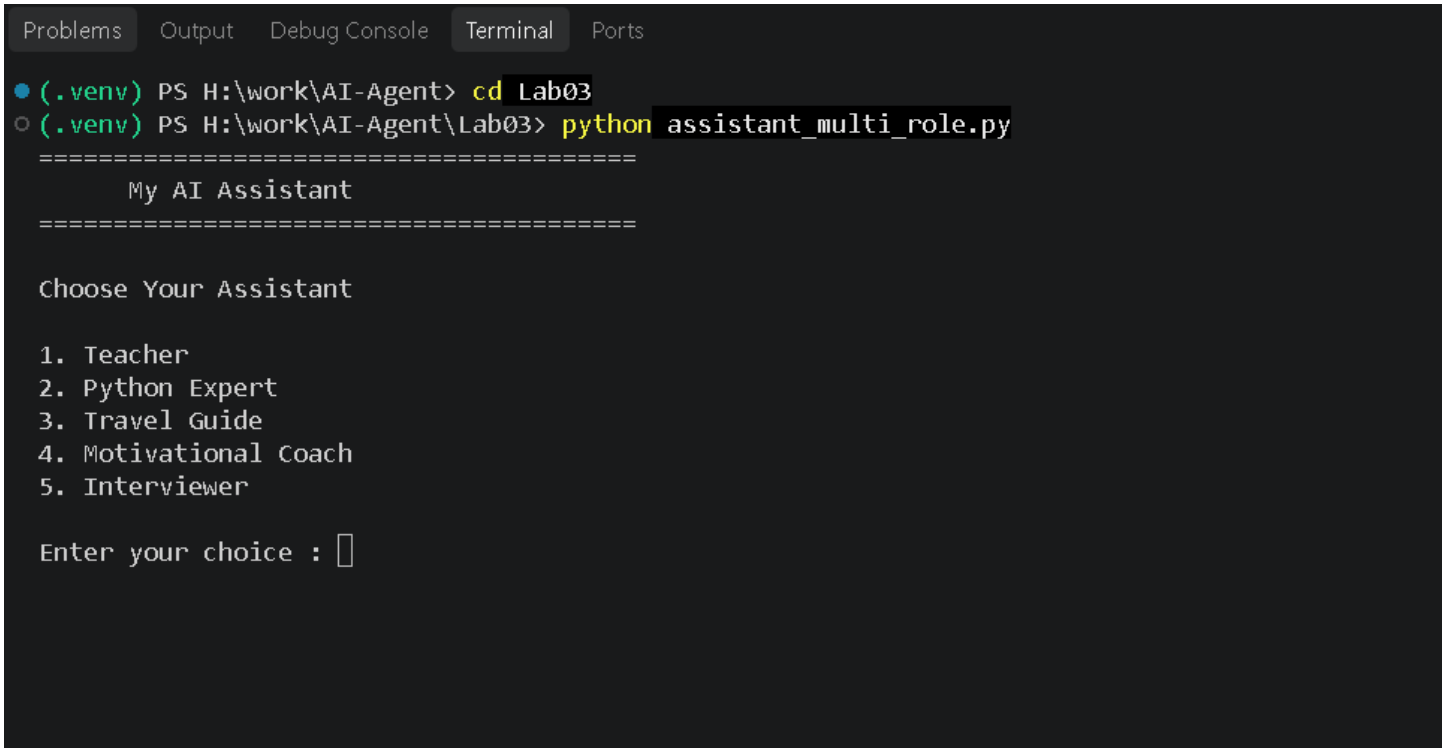
**Real Life Example**:
If you're baking a cake and need to add 10 tablespoons of flour, you can use a loop to add 1 tablespoon each time.

---
```

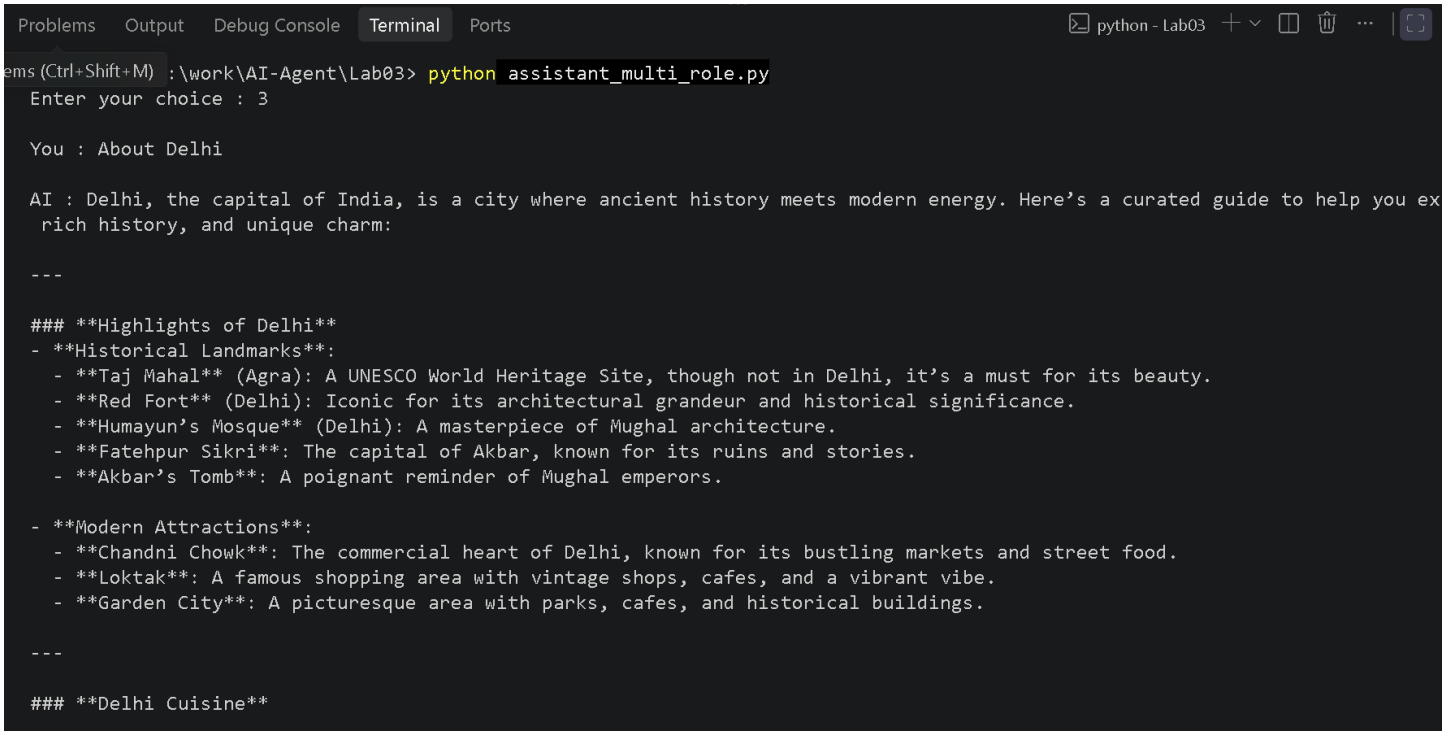
- Now see I ask explain loop it will explain like a teacher with real life example.
- Now we make our assistant for multi role.
- So that in console we select how our assistant behave.



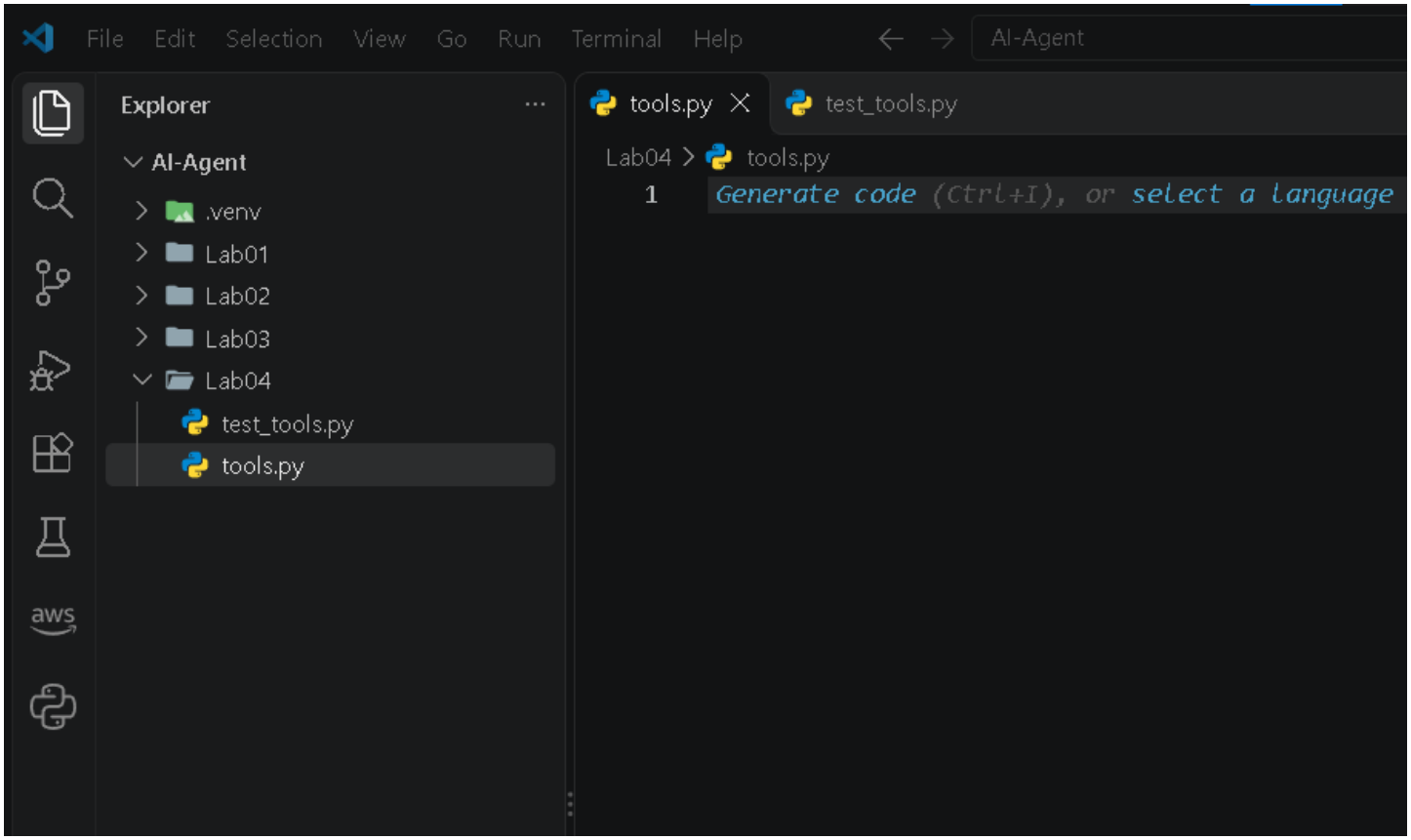
- Now this is the updated code where we create some role list.
- Then we print roles and user give input and then our assistant behaves like that role.
- Code is provided in resource.
- Now see how it works.



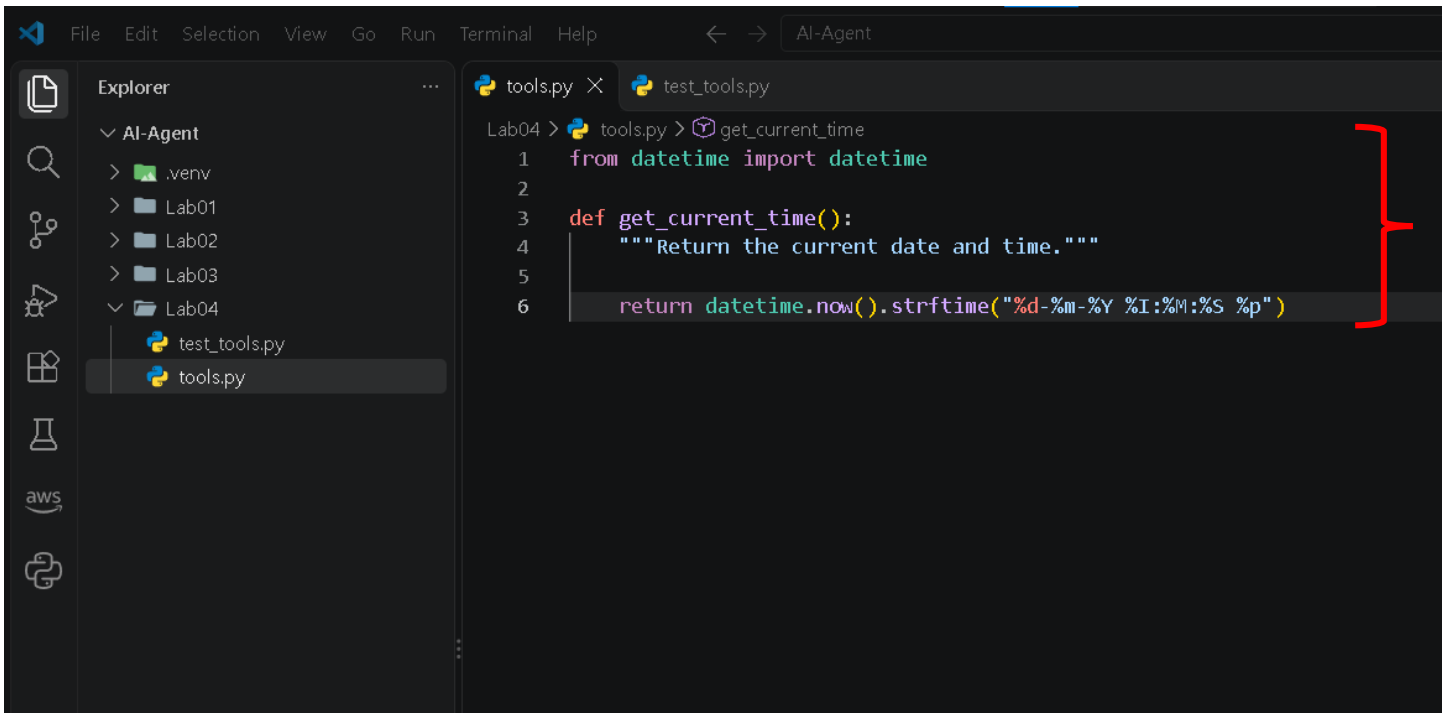
- Now see it ask input how it behaves.
- Like we select 3.



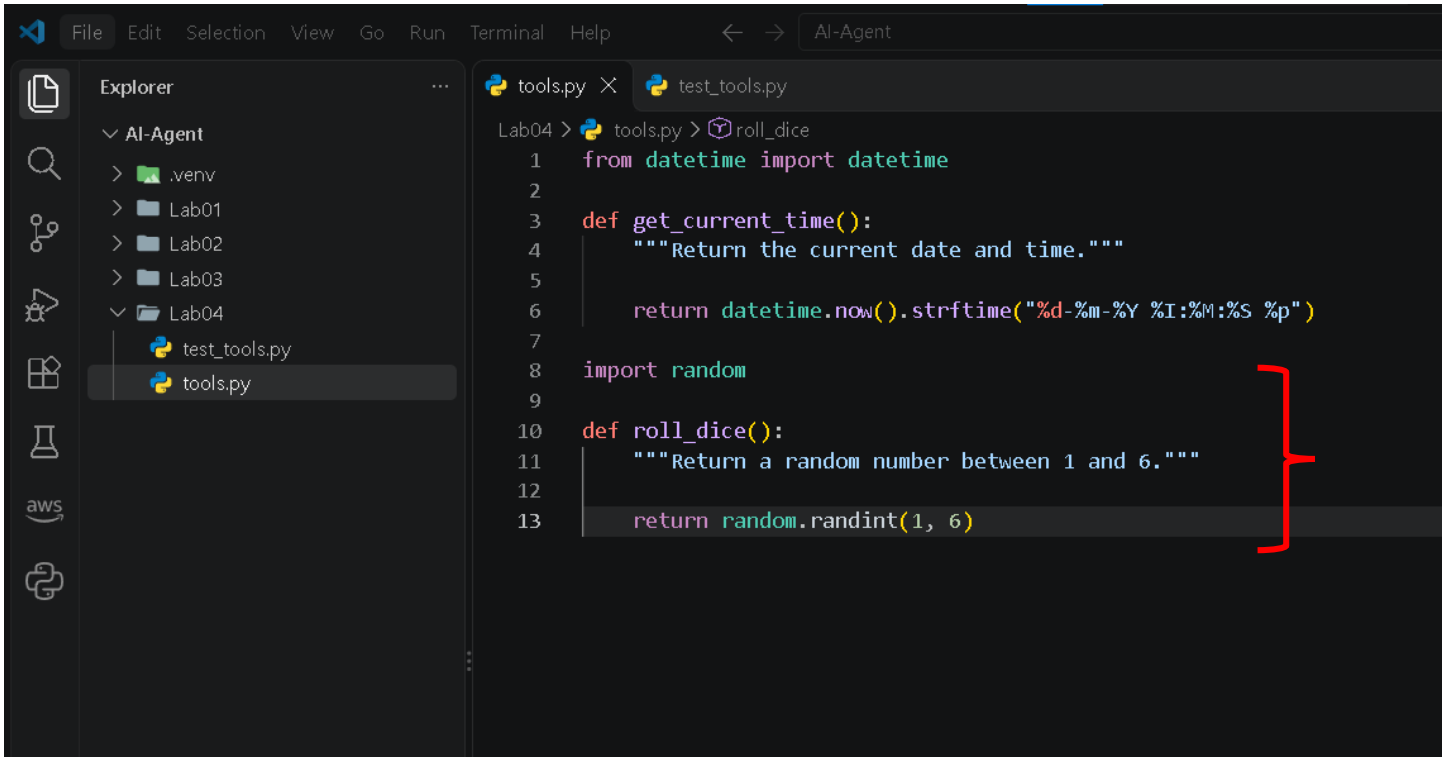
- Now see its behave like a Travel guide.
- Now Let's teach your AI assistant to use tools in Lab04.
- Create a Lab04 folder.



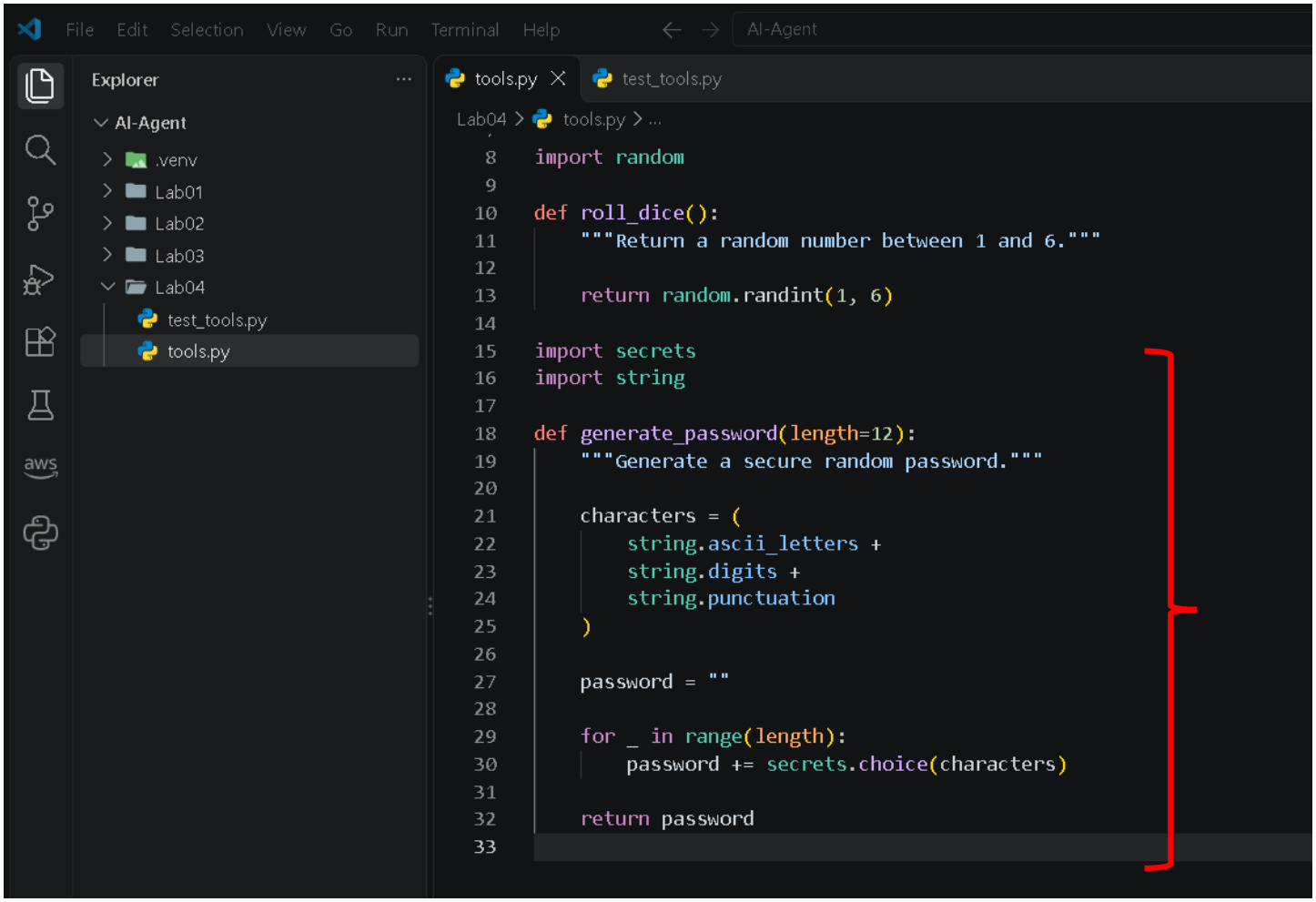
- In Lab)4 create two file test_tools.py and tools.py.
- In tools.py we write out tools code.
- And in our test_tools.py we call our tools function that we write in tools.py.



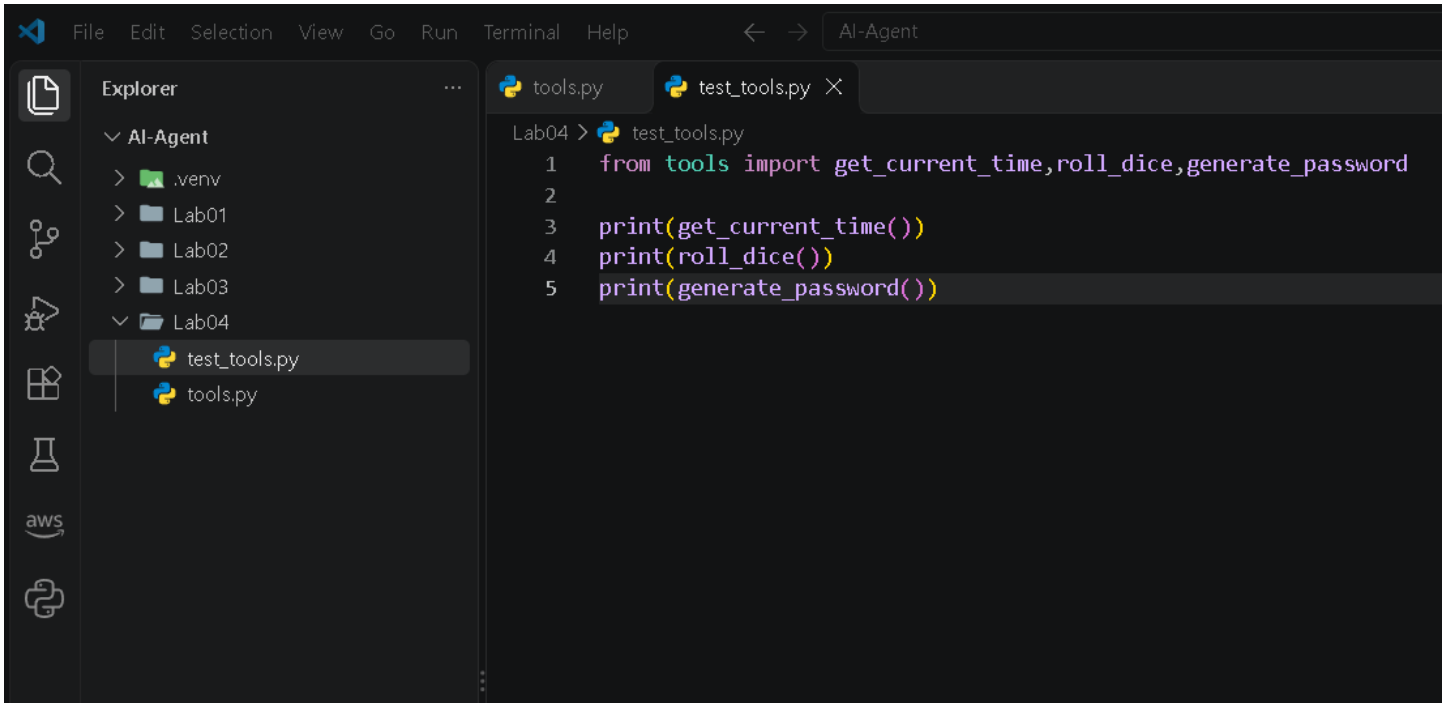
- Our first tool is date and time.
- When we ask date and time it return current date and time from system.



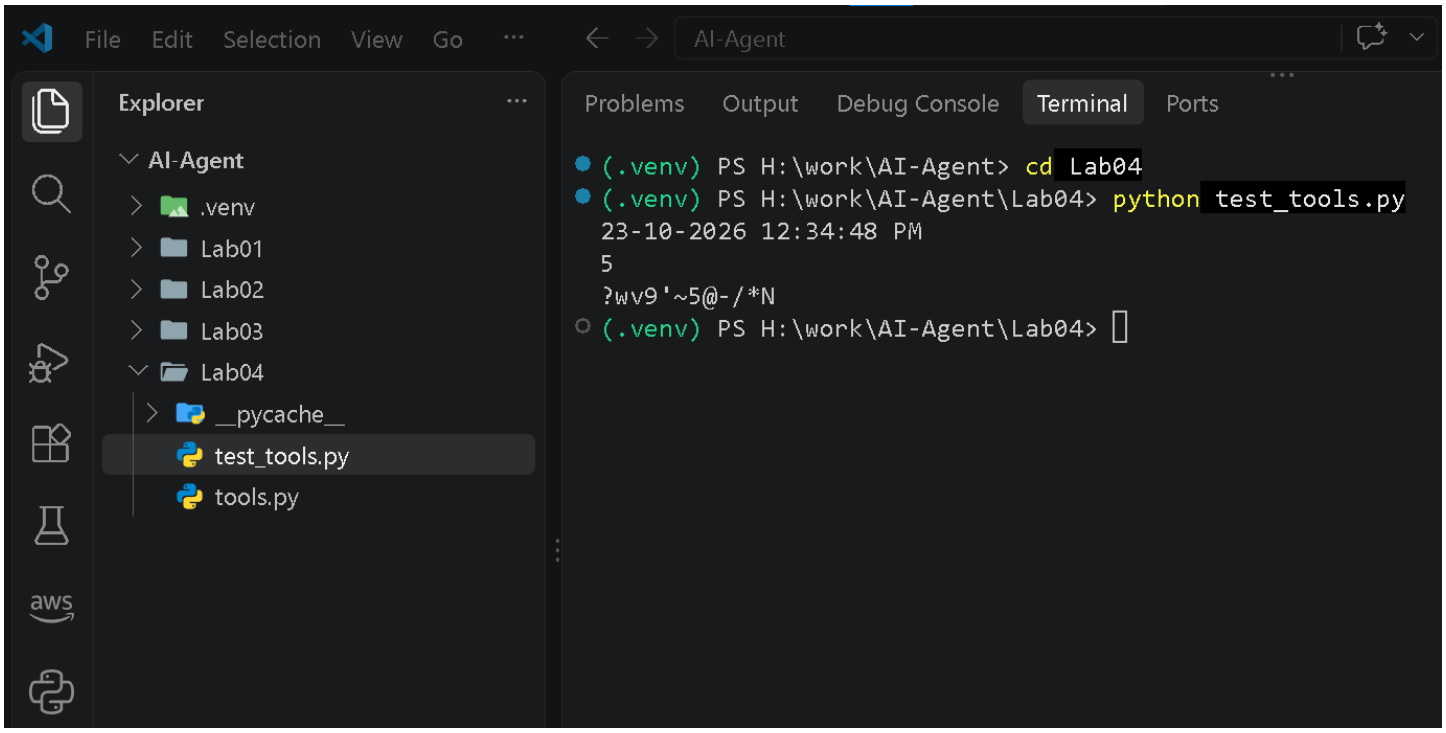
- Now we create second tool that generates random numbers.



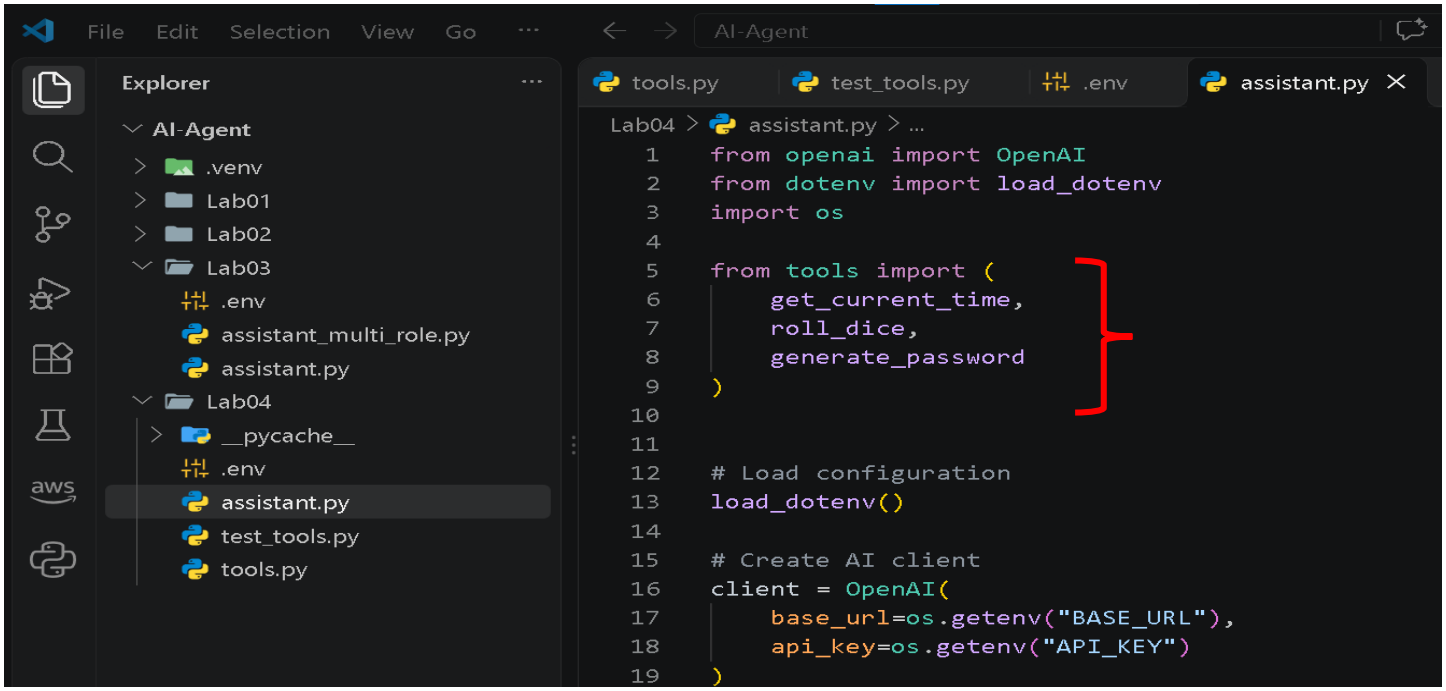
- Now its our third tools that generate password randomly.
- Now open test_tools.py



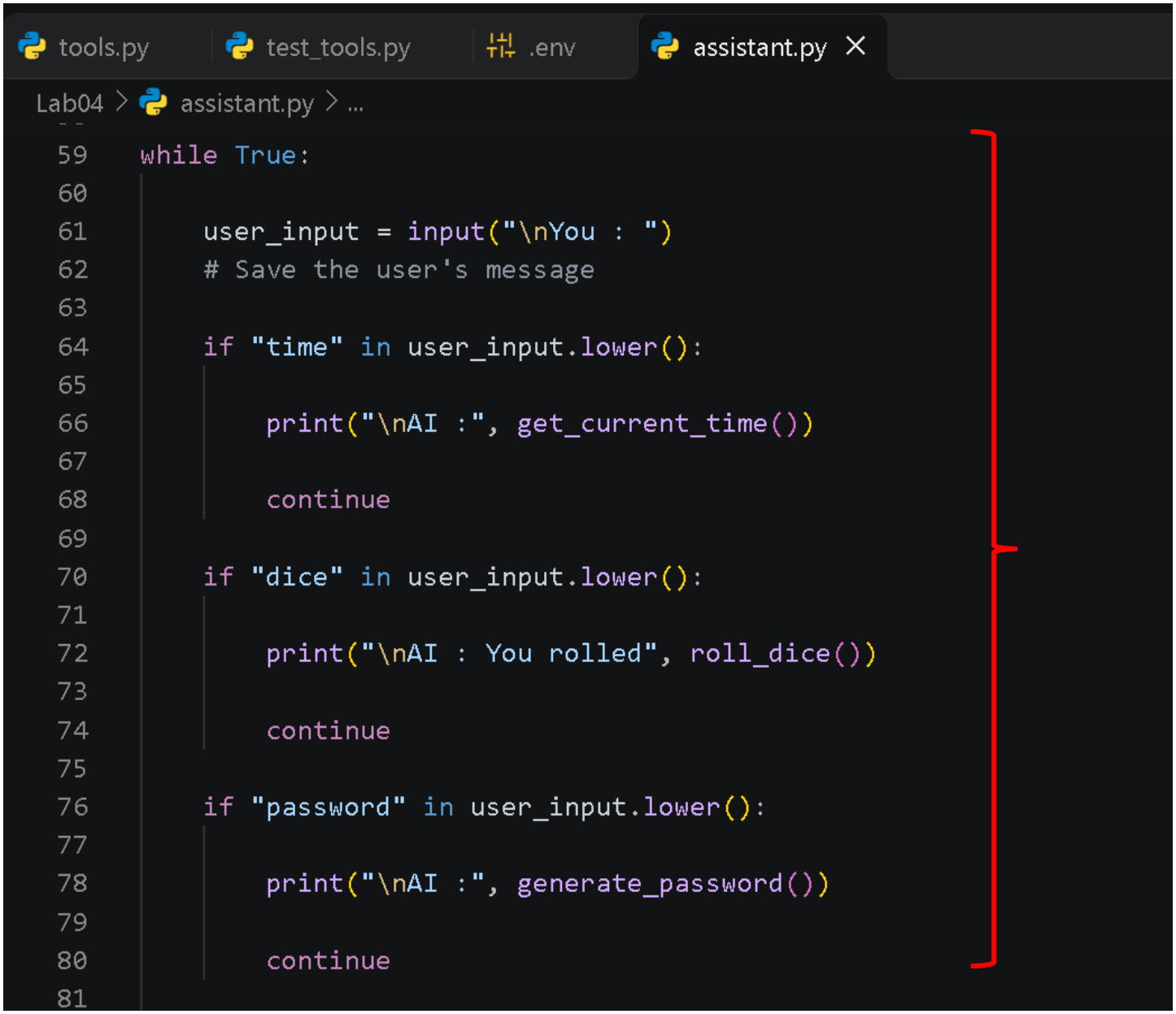
- Now test_tools.py we call our all tools.
- Now let's check its working or not.



- Let's open your terminal and go to Lab04 directory by cd Lab04.
- And then run your test_tools.py file.
- Now see it return date and time, random number, and generate password.
- Our tools are created now our work is that to add tools in our assistant.
- Now we work on assistant file.



- Now I add our assistant.py file and .env file in our Lab04.
- Now I add some codes in it.
- Firstly import our all tools in assistant.py



- Now we add if user give input as time, dice, or password our agent calls our created tools and return output.
- Let's test it.

```
(.venv) PS H:\work\AI-Agent\Lab04> python assistant.py
========================================

Choose Your Assistant

1. Teacher
2. Python Expert
3. Travel Guide
4. Motivational Coach
5. Interviewer

Enter your choice : Teacher

You : what is current time

AI : 23-10-2026 12:51:43 PM

You : Role a dice

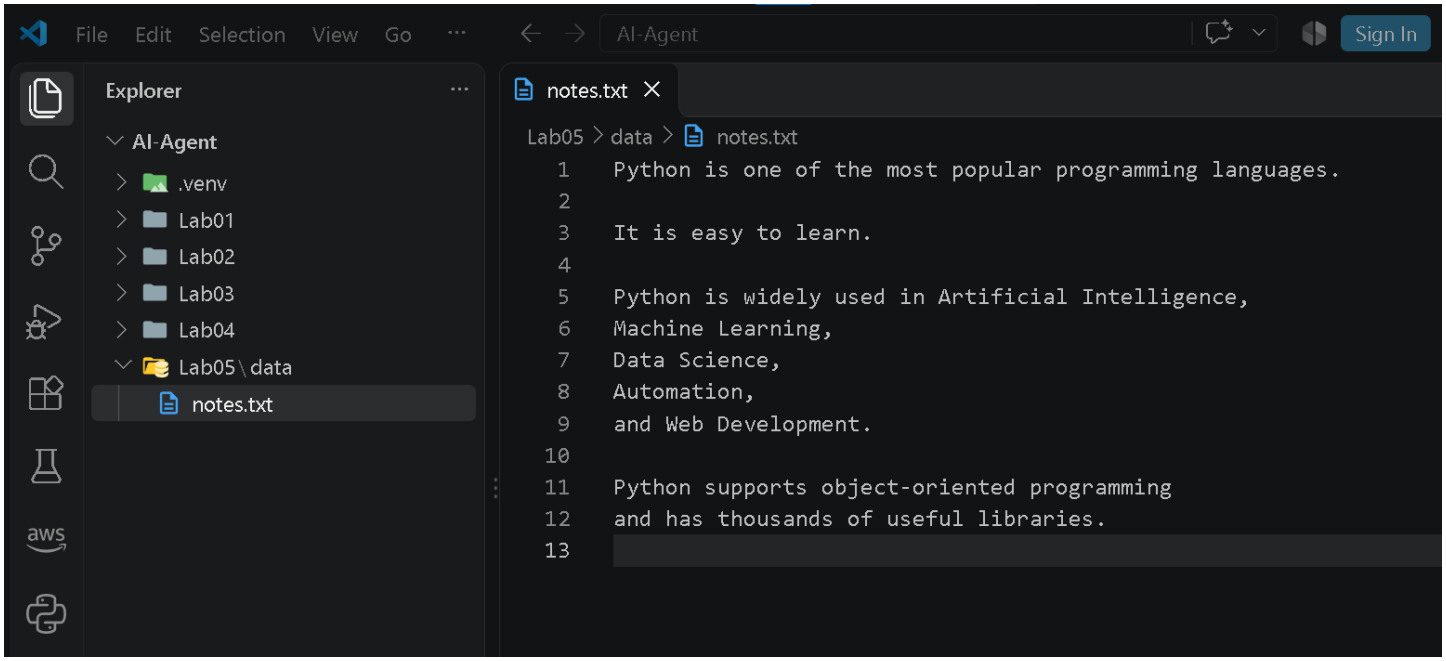
AI : You rolled 6

You : generate a password

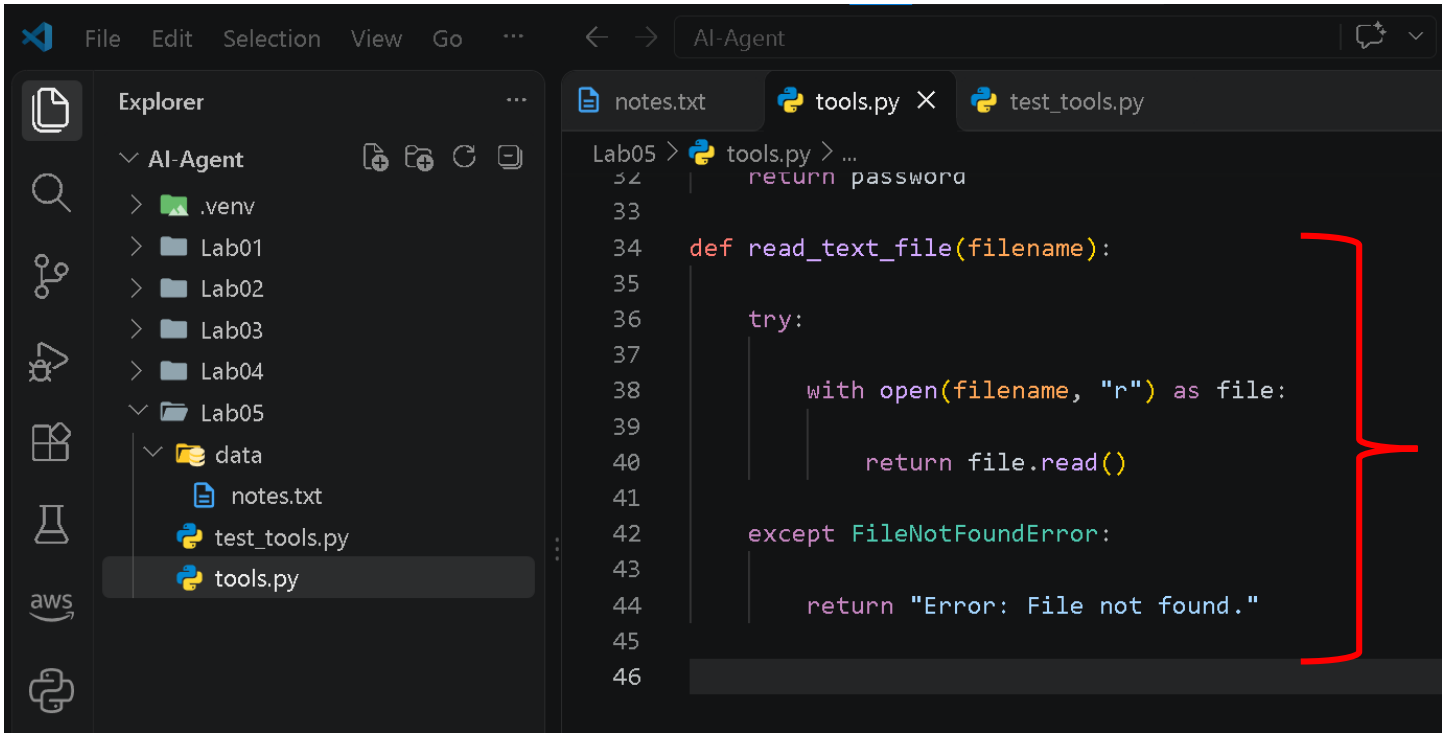
AI : <r.8C%CA*|@a

You :
```

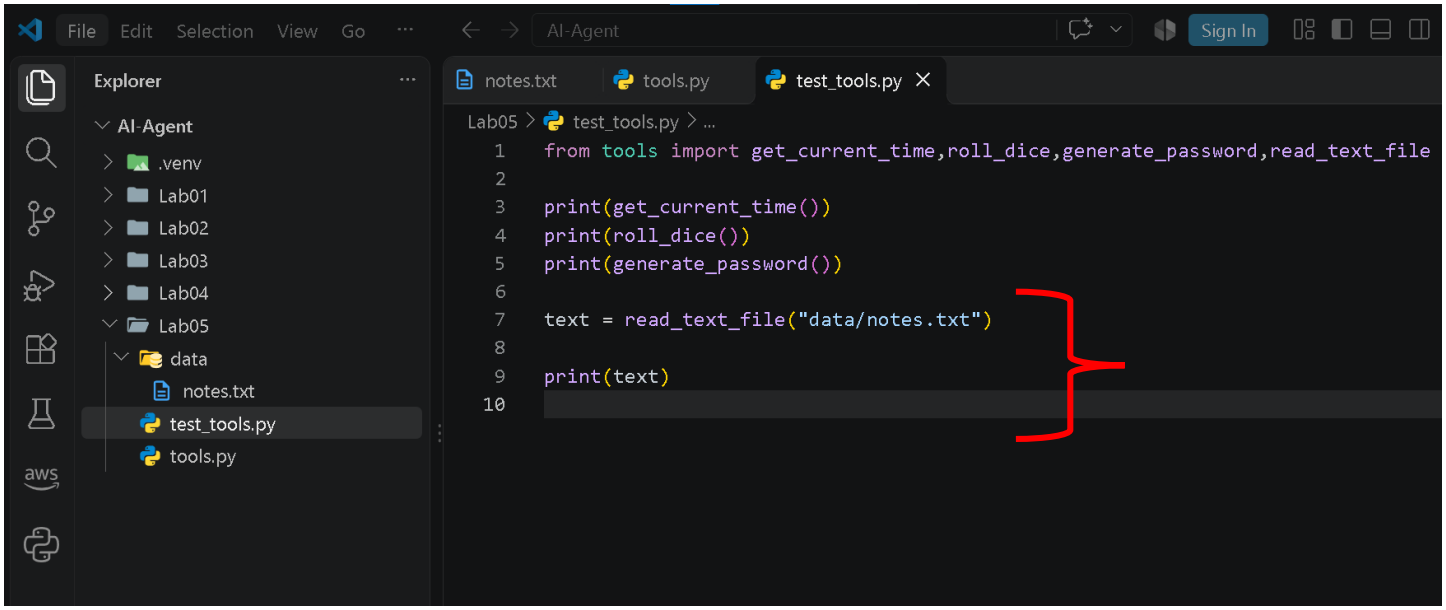
- Now see our assistant doesn't go to AI model.
- Its return output through our tools.
- Now We teach our AI how to read a file.



- Now Let’s create folder Lab05.
- And inside Lab05 create data folder.
- And inside data folder create notes.txt file where we add some readable documents.



- Now add tools.py and test_tools.py file in side Lab05.
- And add this line in our tools.py so that its read file.



- Now import file read_text_file in test_tools.py
- Let’s run its working or not.

```
Problems  Output  Debug Console  Terminal  Ports        powershell - Lab05

(.venv) PS H:\work\AI-Agent> cd Lab05
(.venv) PS H:\work\AI-Agent\Lab05> python test_tools.py
23-10-2026 01:20:59 PM
1
6/`Qtd|qW-y1
Python is one of the most popular programming languages.

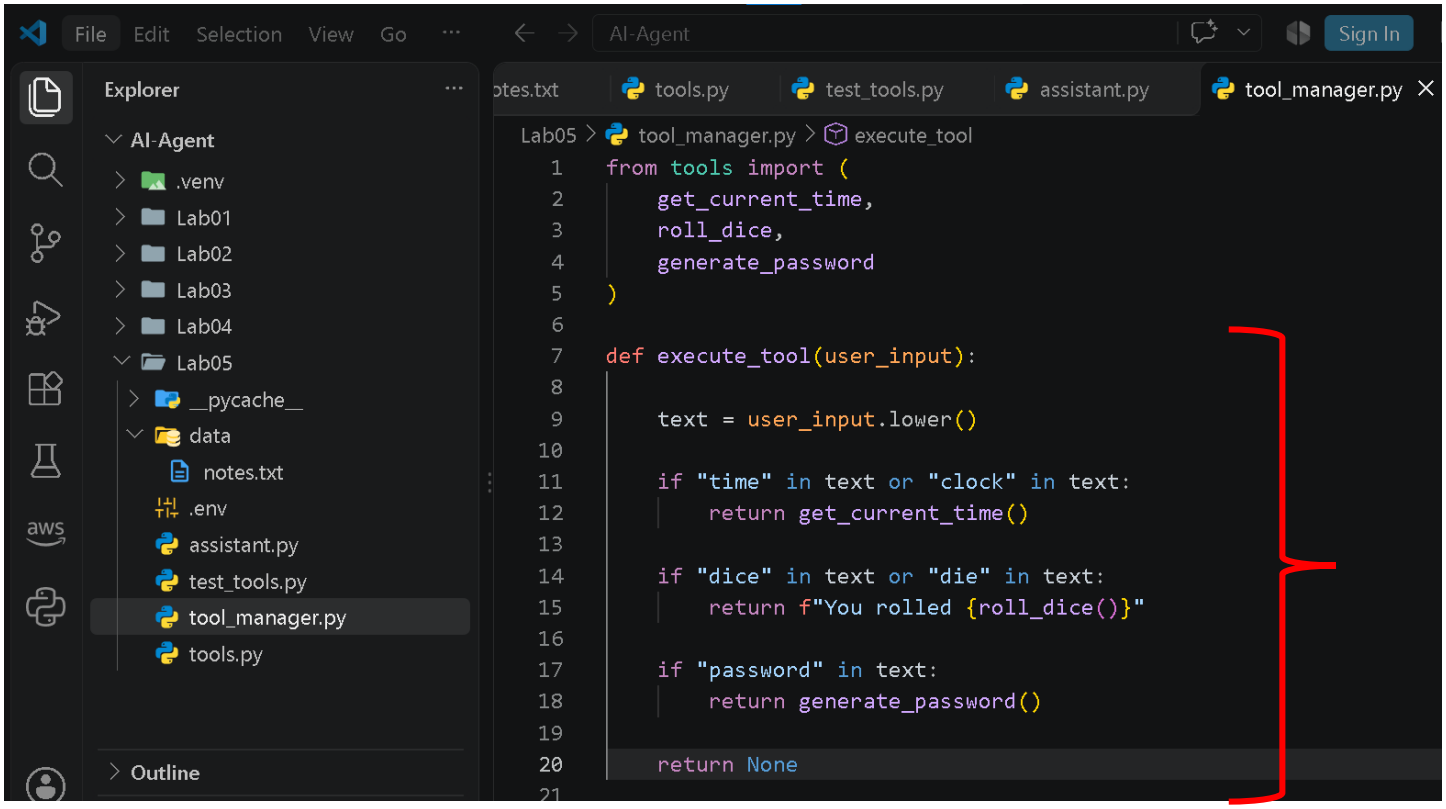
It is easy to learn.

Python is widely used in Artificial Intelligence,
Machine Learning,
Data Science,
Automation,
and Web Development.

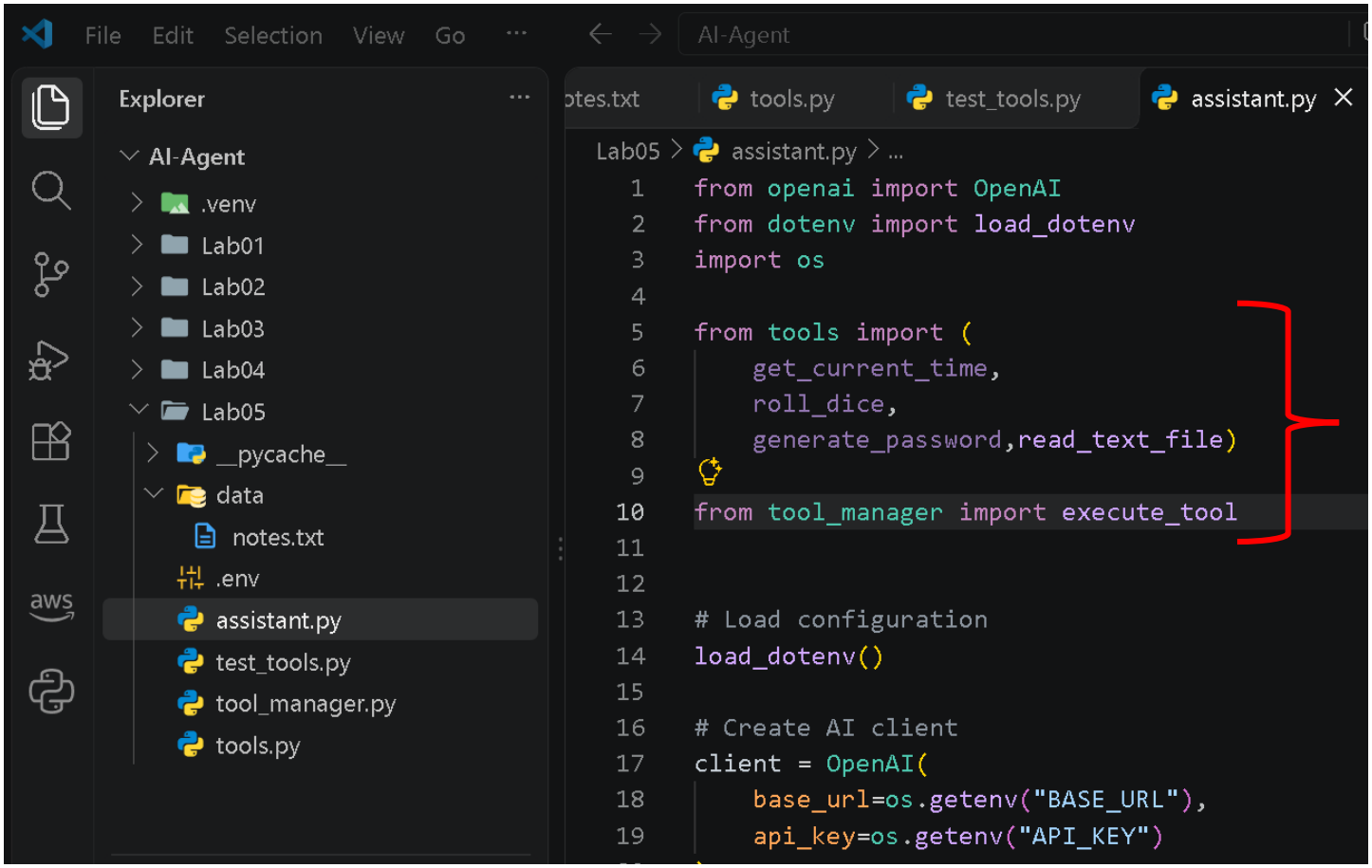
Python supports object-oriented programming
and has thousands of useful libraries.

(.venv) PS H:\work\AI-Agent\Lab05>
```

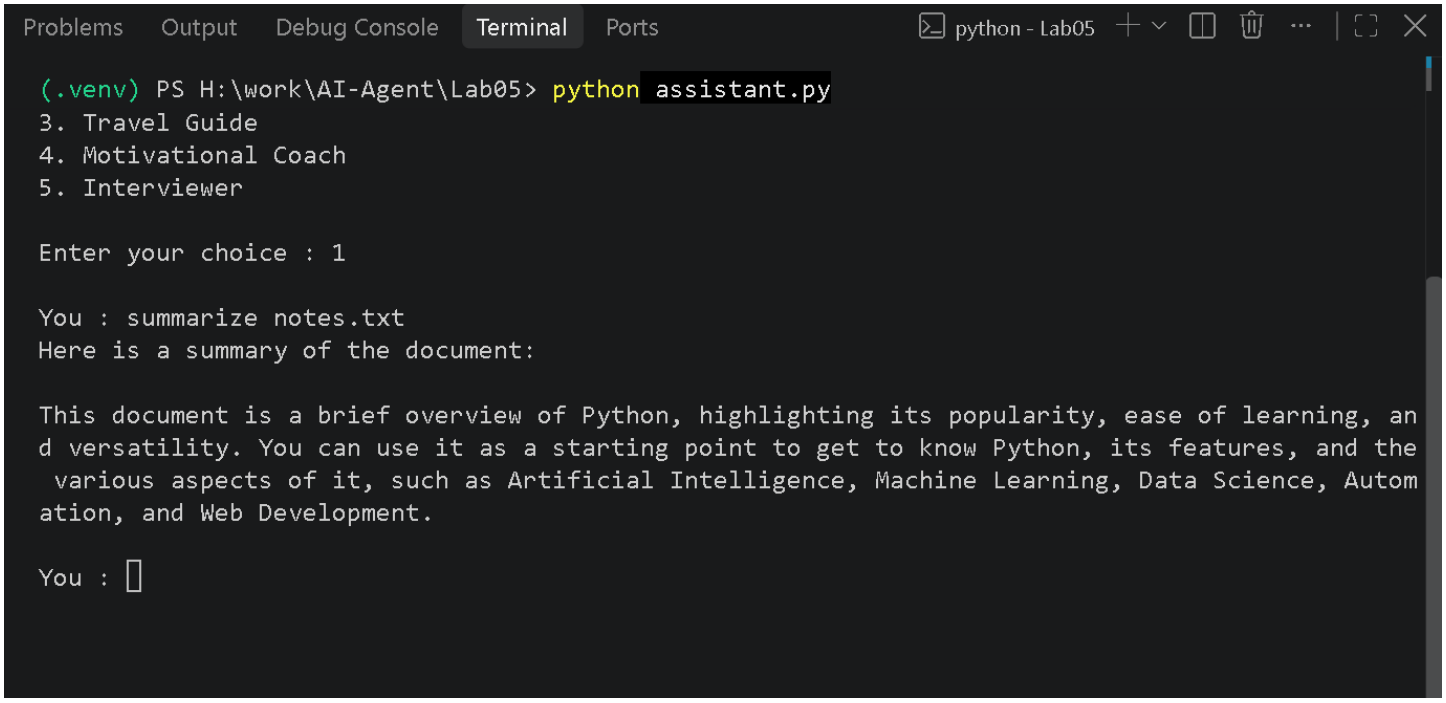
- Now see our tool is read from our files
- Now add this in our assistant so assistant read this file and if I ask any question it reply from this file.



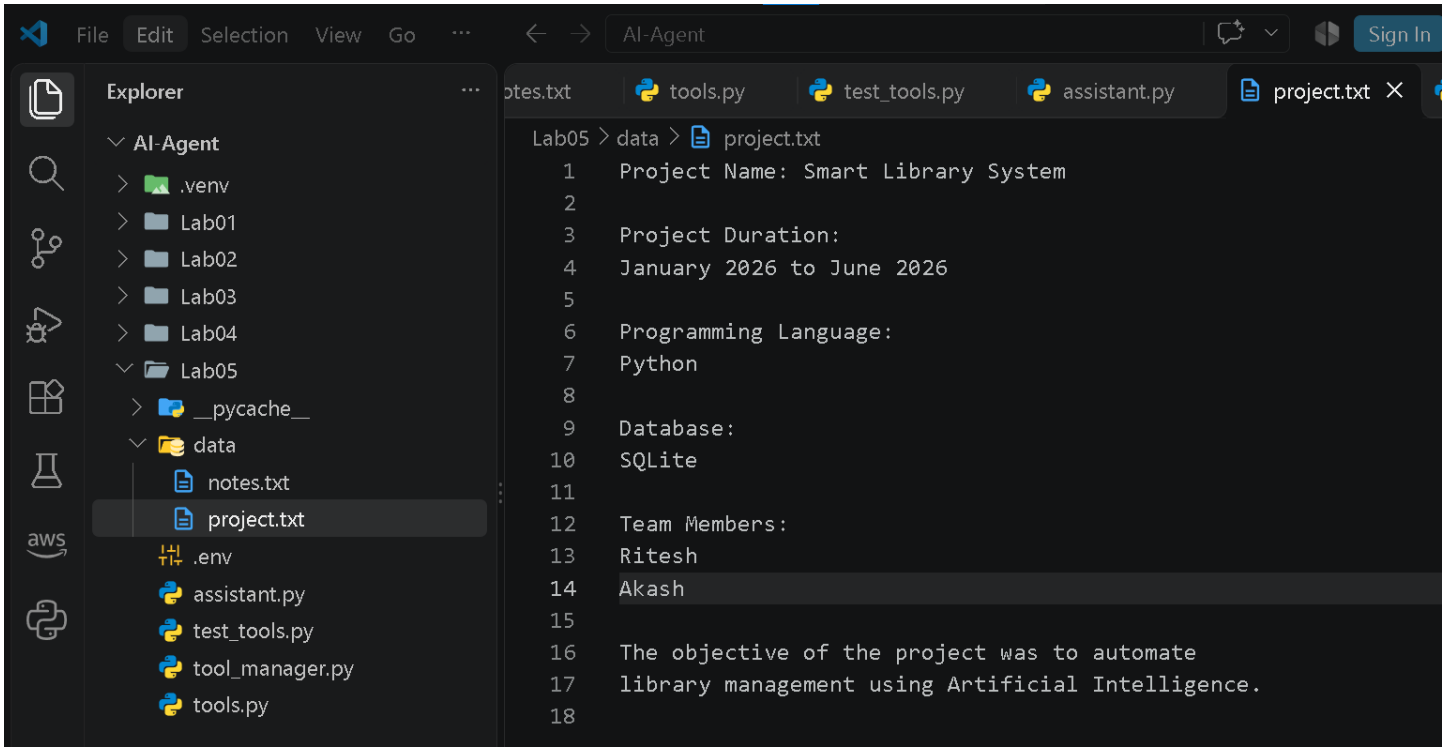
- Now we create tool_manager.py file,
- This file is same as test_tools.py file but here one thing is different that it simple our assistant code.
- Basically that work we done in assistant.py file it makes that code longer so for this we add if else part in tool_manager.py



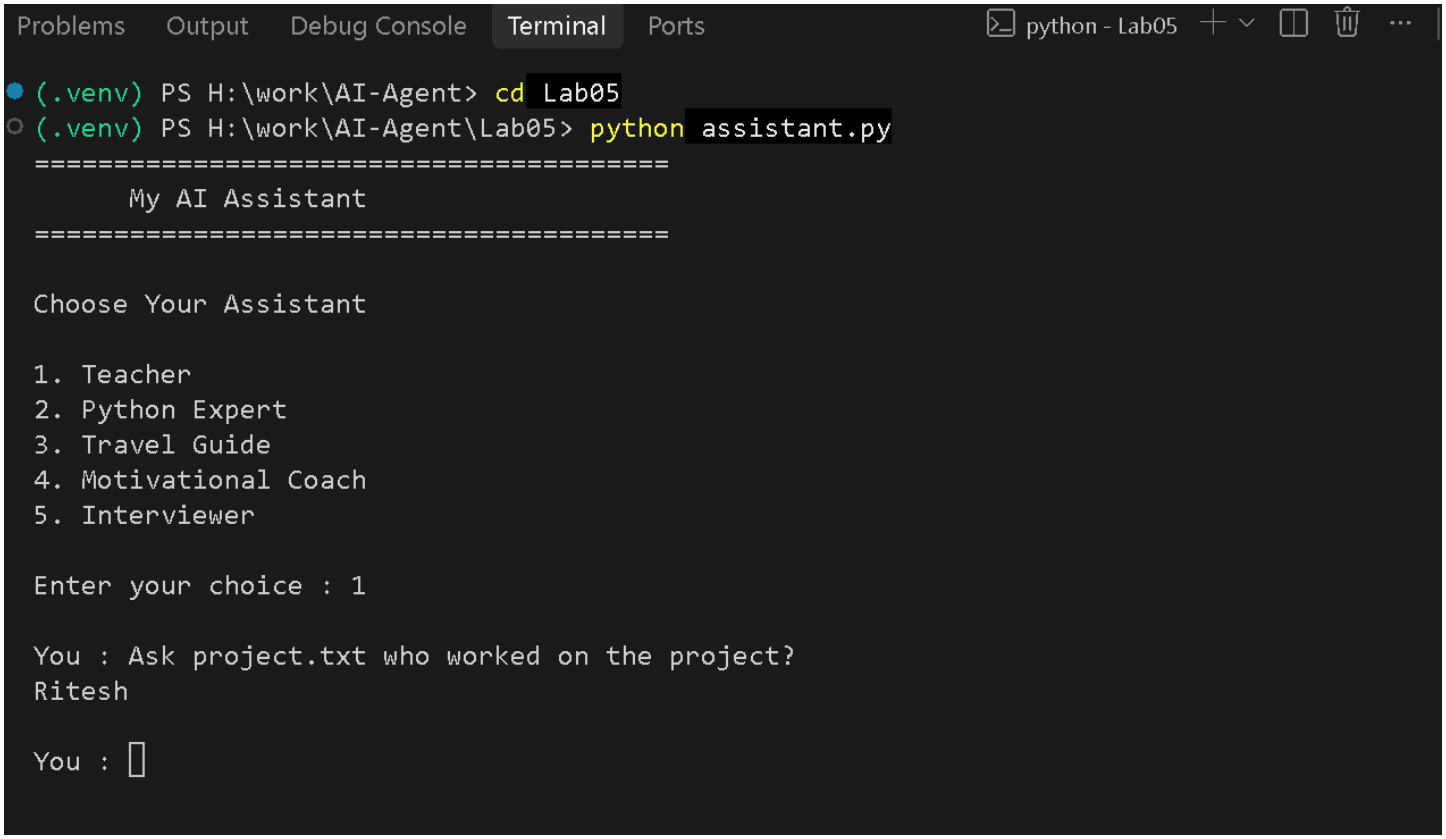
- Now see what I change in assistant.py file.
- Firstly import read_text_file then import our tool_manager.



- Now see our assistant read and reply.
- Now let's check how our assistant Q&A after reading files.



- Create project.txt file inside data folder
- Then ask question.



- Now see its working perfectly